

Hierarchical BERT Classification for Fine-Grained Misinformation Detection: Taxonomy Extension and Extensibility Limits

Submitted in partial fulfilment of the requirements for the degree of

MSc in High-Performance-Computing and Data Science

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Date: 14 August 2026

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Abstract

Misinformation on social media poses significant real-world challenges, yet most automated detection systems reduce a complex and varied problem to a binary decision, real or fake. This framing is insufficient. Different forms of deception operate through different mechanisms, spread for different reasons, and require different responses. A system that cannot distinguish between them cannot meaningfully inform how a platform moderates content, how a journalist investigates a story, or how a policymaker designs regulation. The practical utility of a detection system depends not on whether it can identify that content is deceptive, but on whether it can identify how.

This dissertation develops a hierarchical BERT-based classifier for fine-grained misinformation detection, trained on a large-scale dataset of social media posts labelled across multiple categories of misinformation. Rather than treating all categories as equally unrelated, the model first assigns content to one of three semantic groups reflecting the broad nature of the deception, then makes a specific category prediction within that group. The project further investigates whether the taxonomy can be extended with new misinformation categories through selective retraining of individual classification components, and where that process of extension reaches its practical limit, conducting a formal extensibility experiment using four pre-specified viability criteria to establish the conditions under which extension succeeds and fails.

The hierarchical design consistently outperforms a flat single-stage baseline across the evaluated random seeds, with the largest improvements occurring on minority categories where structural guidance matters most. Two new categories are successfully added, with the principal finding that domain alignment between training data and the target distribution is the decisive factor in extension success, more important than dataset size alone. The formal extensibility experiment finds that text-only classification reaches a natural ceiling determined by the nature of the deception signal rather than data quantity. These findings recommend hierarchical classification for imbalanced fine-grained detection tasks, domain alignment as a design constraint rather than an afterthought, and multimodal extension as the necessary next step for categories whose deception is carried through visual rather than textual content.

Chapter 1. Introduction

Misinformation is not a new problem, but the scale at which it now operates is unprecedented. The internet has removed the gatekeeping function that editors, publishers, and broadcasters once provided: the filters that slowed the spread of false information long enough for corrections to catch up. What replaced those filters are recommendation algorithms optimized for engagement rather than accuracy, and social networks that allow a single false story to reach millions of people in minutes. More recently, a new dimension has emerged that makes the problem significantly harder: Artificial intelligence can now generate convincing misinformation at scale, faster than any human fact-checker can review it. Despite this, the dominant approach in automated detection remains binary (real or fake), a framing that conflates fundamentally different forms of deception and limits the practical utility of any system built on it. This chapter establishes the problem this dissertation addresses, explains why the approach taken here differs from most existing work, and introduces the core ideas at a conceptual level before the technical detail begins in Chapter 3.

1.1 The World We Live In

The consequences of misinformation are well documented. During the COVID-19 pandemic, false information about vaccines spread so rapidly that the World Health Organisation declared a simultaneous infodemic alongside the medical one. In the 2016 and 2020 United States elections, fabricated stories about candidates were shared more widely than verified reporting from established outlets. In Myanmar in 2017, inflammatory false content spread on social media contributed directly to ethnic violence against the Rohingya people, violence that the United Nations later described as having been facilitated by the platform's algorithmic amplification of hate speech.

These cases are not exceptional. They are the predictable consequence of a media environment where content is amplified by engagement rather than accuracy, and where the volume of posts is far beyond what human reviewers can assess. The research community has responded by developing automated misinformation detection systems, machine learning models that can assess content at scale. This dissertation contributes to that effort, but from a different starting premise: that the type of deception matters, and that a system which cannot distinguish between types cannot meaningfully act on what it finds.

1.2 The Limitation of Existing Approaches

Most automated detection systems treat misinformation as a binary problem: is this content real or fake? This framing is insufficient. A misattributed photograph, a satirical article, and a fabricated news report are all labelled identically under a binary classifier, yet they operate through different mechanisms, spread for different reasons, and require different responses from platforms, journalists, and policymakers. The practical value of a detection system depends not on whether it can identify that something is deceptive, but on whether it can identify how, and binary classification, by design, cannot answer that question.

Binary classification dominated early misinformation research for understandable reasons: binary labels are easier to collect at scale, binary models are simpler to train and evaluate, and the first priority was demonstrating that automated detection was feasible at all. Having established feasibility, the field is now in a position to ask a more precise question: not whether a piece of content is deceptive, but in what way.

1.3 What This Project Does and Why This Matters in Practice

This project develops a system that classifies misinformation across multiple distinct categories, each corresponding to a different form of deception, using a hierarchical two-stage architecture built on BERT. A coarse classification stage first assigns content to a broad semantic group reflecting the nature of the deception, and a fine classification stage then identifies the specific category within that group. Two new misinformation categories are subsequently added to the taxonomy using a selective retraining strategy that does not require rebuilding the entire model, and the project formally investigates how far this extension process can go, establishing empirically where the system can no longer learn to distinguish new categories reliably and why. To the author's knowledge, no prior published work has applied hierarchical classification to fine-grained misinformation detection on this dataset, nor formally investigated the extensibility limits of such a system, and it is this combination of contributions that distinguishes the project from existing work in the field. The full technical design is described in Chapter 3.

A system that identifies the type of misinformation rather than simply its presence has concrete practical value, because the type determines the appropriate response. Different category types warrant different responses: satire requires a context label rather than removal, imposter content requires source investigation, and manipulated media requires referral to image forensics. A binary label cannot support any of these workflows. The same argument applies beyond content moderation, where researchers need category-level data to ask meaningful empirical questions, policymakers need to distinguish between forms of deception that are illegal, harmful but legal, or protected expression, and journalists need to know what kind of investigation a story requires before they begin. None of these distinctions are possible from a binary label alone.

A further dimension motivates the work. The misinformation landscape evolves continuously. AI generated content, for instance, was not a recognised category five years ago. A detection system that requires full retraining every time a new category emerges will be unable to keep pace with an evolving problem. This project investigates whether a hierarchical design can be extended incrementally, and where that incremental extension reaches its limit.

1.4 Research Questions

Three research questions structure this dissertation:

RQ1: Does a hierarchical two-stage classifier outperform a flat single-stage classifier for fine-grained misinformation detection? The hierarchical design introduces architectural complexity that is only justified if it produces measurably better results than the simpler alternative under controlled conditions. Without a positive answer, the architectural complexity of the hierarchical design is unjustified. This question is addressed in Chapter 4.

RQ2: Can the taxonomy be extended with new categories using selective retraining, without degrading performance on existing categories? A detection system that cannot grow as the misinformation landscape evolves will become obsolete. This question tests whether the selective retraining strategy works in practice and is addressed in Chapters 5 and 6.

RQ3: At what point does taxonomy extension stop being beneficial, and what are the failure modes when it does? This question produces the most empirically original contribution of the project. It requires not just training models but designing a formal evaluation framework that can distinguish genuine learning from superficial pattern matching and identify the mechanistic reason for failure when it occurs. This question is addressed in Chapter 8.

Chapter 2. Background and Literature Review

A comprehensive literature review was conducted during the Project Preparation phase and is documented in the accompanying feasibility report. Rather than reproducing that review in full, this chapter provides a structured summary of its key findings, identifies the gaps that motivate this project, and notes four areas of prior work that were not covered in the feasibility report but are directly relevant to the experimental contributions of this dissertation.

2.1 Summary of Prior Work

The feasibility report reviewed five bodies of literature. The key findings are summarized below:

Fake news detection landscape. Early approaches relied on hand-crafted linguistic and network features (Castillo et al., 2011; Rubin et al., 2016), later superseded by CNNs and RNNs (Wang, 2017; Ruchansky et al., 2017). BERT-based classifiers represent the current state of the art (Kula et al., 2021). The overwhelming majority of published systems treat misinformation detection as binary classification, collapsing fundamentally different forms of deception into a single label.

Available datasets. Most benchmarks are either binary (FakeNewsNet, CRED BANK) or use truthfulness grades rather than deception type labels (LIAR). Fakeddit (Nakamura et al., 2020) is the largest benchmark providing six category labels derived from First Draft's professional typology (Wardle, 2017) but is significantly class-imbalanced and has not previously been approached with a hierarchical classifier.

Transformer models and transfer learning. The Transformer (Vaswani et al., 2017) introduced self-attention for long-range dependencies. BERT (Devlin et al., 2019) extended this with bidirectional masked language modelling, producing richer contextual representations. Fine-tuning pre-trained BERT consistently outperforms task-specific architectures trained from scratch. bert-base-uncased was selected over RoBERTa and DeBERTa to maintain comparability with prior work.

Hierarchical classification. Silla and Freitas (2011) identify three main approaches, of which local classifiers per level is adopted here. Kowsari et al. (2019) demonstrated consistent improvements over flat baselines, largest for semantically similar categories. Rashkin et al. (2017) found broad deception categories more learnable than fine-grained ones, motivating a coarse-to-fine architecture.

Catastrophic forgetting and selective freezing. McCloskey and Cohen (1989) documented catastrophic forgetting, whereby gradient updates for a new task overwrite knowledge of previous tasks. Kirkpatrick et al. (2017) formalised this as a fundamental property of gradient-based learning. Selective freezing, holding previously learned components fixed while updating only those needed for the new task, is adopted here as a practical alternative to more complex continual learning approaches.

2.2 Gaps Identified

Three gaps in the existing literature motivate this project:

Gap 1: No hierarchical approach on fine-grained misinformation benchmarks. Despite the well-established benefits of hierarchical classification for imbalanced multi-class problems, no prior work has applied this approach to a fine-grained misinformation dataset with a semantically structured

label space. The grouping of misinformation categories by their underlying deception mechanism has not been exploited in any prior architecture evaluated on such benchmarks.

Gap 2: No formal study of taxonomy extensibility. The misinformation literature contains no formal investigation of whether a trained classifier can be extended to cover new categories without full retraining. Given that the misinformation landscape evolves continuously, this is a practically significant gap. This project provides the first empirical answer, including a formal viability framework with pre-specified criteria.

Gap 3: No domain alignment framework for category extension. The iterative experiments in Chapter 5 reveal that the match between training data format and deployment data format is the decisive factor in extension success, more important than dataset size alone. This finding is not documented in any prior work on taxonomy extension or continual learning in the misinformation domain.

2.3 Additional Literature

Four areas of prior work not covered in the feasibility report are directly relevant to the experimental contributions of this dissertation. Each is summarized below with an explicit connection to the chapter where it applies.

Clickbait detection. Chen et al. (2015) framed clickbait as a form of false news, arguing that deliberately misleading framing constitutes deception even when the underlying facts are accurate. The deceptive signal is carried in headline style rather than factual inaccuracy, making it detectable from title text alone. Ahmad et al. (2020) showed BERT consistently outperforms shallower models on clickbait classification. Cao et al. demonstrated that high overall accuracy could mask poor minority-class performance, directly informing the evaluation strategy in Chapter 6.

Class imbalance in hierarchical classification. Babbar et al. (2016) found that hierarchical classifiers outperform flat classifiers on imbalanced data because the coarse stage concentrates each fine head's capacity on a more balanced subset, while flat classifiers allow majority classes to dominate the loss. This directly explains the Chapter 4 pattern where the largest gains appear on Satire and Imposter Content, precisely where theory predicts the hierarchical advantage to be greatest.

Alternative approaches to class imbalance. Re-weighted cross-entropy, class-balanced loss, and focal loss address imbalance through the loss function rather than through label-space structure, making them complementary rather than equivalent to hierarchical classification. The initial experiments deliberately use standard cross-entropy to isolate the architectural effect of the hierarchy. Loss re-weighting is introduced only after the error analysis in Chapter 7 identifies minority-category starvation as a specific failure mode.

Domain alignment and continual learning. Ben-David et al. (2010) established that fine-tuning performance degrades in proportion to the divergence between training and deployment distributions and that additional samples cannot compensate when distributions differ sufficiently, explaining the domain mismatch failures in Chapter 5. Lopez-Paz and Ranzato (2017) showed that selective parameter freezing is theoretically optimal when a new task shares a representational basis with existing ones, justifying the selective freezing strategy and framing the extensibility limit in Chapter 8 as a test of where this condition breaks down.

Model calibration. Guo et al. (2017) established Expected Calibration Error as the standard measure of the gap between stated confidence and empirical accuracy and demonstrated that label smoothing is an effective mitigation for systematic overconfidence. Both findings are directly relevant: overconfident wrong predictions are identified as a significant failure mode in Chapter 7, motivating label smoothing in V5 retraining, and ECE is adopted as a formal viability criterion in Chapter 8.

Chapter 3. Methodology

This chapter describes the complete technical design of the system built in this project, with a justification for each architectural decision, training choice, and evaluation metric. The goal is to document not just what was done but why each decision was made over the available alternatives, so that the reasoning is transparent and the experimental setup is reproducible. The experiments that use this system are described in Chapters 4 through 8.

3.1 System Overview

The system is a two-stage hierarchical text classifier for fine-grained misinformation detection. It takes as input a short text string, typically a social media post title of eight to twelve words, and produces as output one of eight misinformation category labels, each corresponding to a distinct form of deception. The dataset used to train and evaluate the system is described in Section 3.2.

The classification proceeds in two sequential stages. In Stage 1, a shared BERT encoder converts the input title into a 768-dimensional representation, which is passed to a coarse classification head. The coarse head assigns the post to one of three semantic groups: Authentic, Structural Deception, or Fabricated/Manipulated. In Stage 2, the same BERT representation is passed to the fine classification head for the predicted group, which makes the specific category prediction within that group.

The two stages share the same BERT encoder but have entirely independent classification heads. This design means that when a new category is added to one group, only that group's fine head and the coarse head need to be retrained. The BERT encoder and all other fine heads remain frozen, making taxonomy extension computationally practical. This property is tested directly in Chapter 5, 6, and 8.

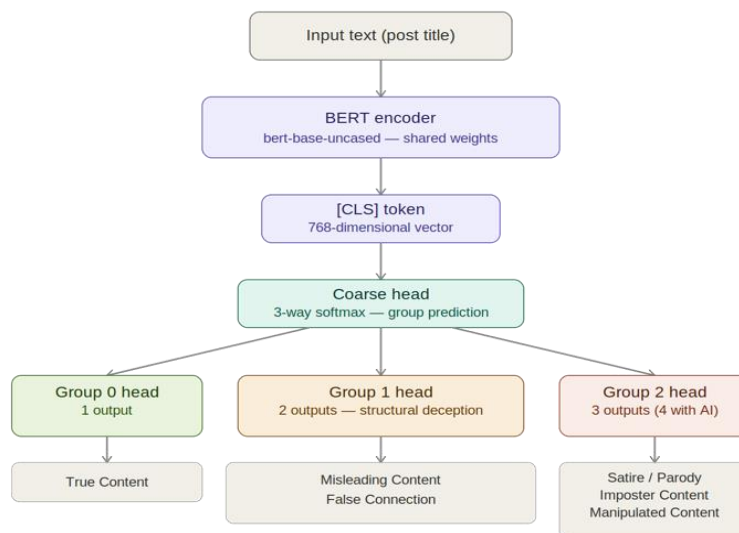


Figure 1: Hierarchical architecture.

3.2 Dataset

The dataset used in this project is Fakeddit (Nakamura, Levy, and Wang, 2020), a large-scale benchmark for fine-grained misinformation detection built from Reddit. Subreddit membership provides a strong and consistent proxy for content type, allowing labels to be assigned systematically and at scale without requiring individual human annotation of every post.

3.2.1 Why Fakeddit Over Other Datasets:

Several alternative datasets were considered before Fakeddit was selected. The choice was driven by the specific requirements of this project: a large-scale, multi-class dataset with a principled label taxonomy grounded in professional practice.

LIAR (Wang, 2017) has two limitations for this project: its labels represent degrees of truthfulness rather than types of deception and do not map onto the semantic groupings that motivate the hierarchical design, and at 12,500 samples it is an order of magnitude smaller than required, which would limit statistical reliability and worsen the class imbalance problem.

FakeNewsNet (Shu et al., 2020) contains articles from PolitiFact and GossipCop labelled as real or fake. Like most benchmarks in this space, it is binary and does not distinguish between types of misinformation at all. It is therefore unsuitable for the core research question of this project, which requires fine-grained category labels.

CREDBANK (Mitra and Gilbert, 2015) focuses on rumour detection on Twitter, with labels indicating credibility rather than misinformation type. Again, binary and at a different granularity than required.

The key properties that make Fakeddit the right choice for this project are:

Property	Fakeddit	LIAR	FakeNewsNet
Scale	682,000 posts	12,500 statements	23,000 articles
Label type	6 misinformation types	6 truthfulness grades	Binary real/fake
Grounded in professional typology	Yes (First Draft)	No	No
Supports hierarchical grouping	Yes	No	No
Social media format	Yes (Reddit titles)	No (Political statements)	No, full articles

Table 1: Dataset Options

Fakeddit is the only publicly available dataset that simultaneously provides the scale, the categorical label depth, and the social media format required to answer the research questions in this project. No other dataset supports the hierarchical taxonomy design described in this chapter.

3.2.2 Dataset Splits:

The dataset is divided into three parts:

Split	Samples	Purpose
Training	564,000	Model training
Validation	59,342	Hyperparameter tuning and early stopping
Test	59,319	Final evaluation, used only once

Table 2: Dataset Splits

The test set is used only once, at the end of each experiment, to produce the final reported results. Using it during development would constitute data leakage, causing reported performance to overestimate real-world generalisation. All hyperparameter decisions are made using the validation set only.

3.2.3 Label Structure and Class Imbalance:

Fakeddit assigns each post one of six labels derived from First Draft's professional misinformation typology:

Label	Category	Nature of deception
0	True Content	No deception
1	Satire/Parody	Fiction presented as news
2	Misleading Content	Real event, false context
3	Imposter Content	Fake source impersonating real one
4	False Connection	Real image or headline, mismatched
5	Manipulated Content	Doctored image or video

Table 3: Fakeddit Dataset Structure

These six labels are remapped in this project into a hierarchical structure with three groups:

- Group 0: Authentic - True Content (label 0)
- Group 1: Structural Deception Misleading Content (label 2) False Connection (label 4)
- Group 2: Fabricated/Manipulated Satire/Parody (label 1) Imposter Content (label 3) Manipulated Content (label 5)

The grouping reflects the fundamental nature of the deception in each category. Group 0 contains genuinely accurate content with no deception present. Group 1 contains content where something real is presented in a misleading structural context, the event happened, the image exists, but it has been framed or paired incorrectly. Group 2 contains content that is fabricated or significantly altered at source, the content itself has been invented or manipulated. This grouping is grounded in the misinformation typology literature and reflects semantic relationships that prior work on hierarchical classification has shown to be learnable (Silla and Freitas, 2011).

The training set distribution is as follows:

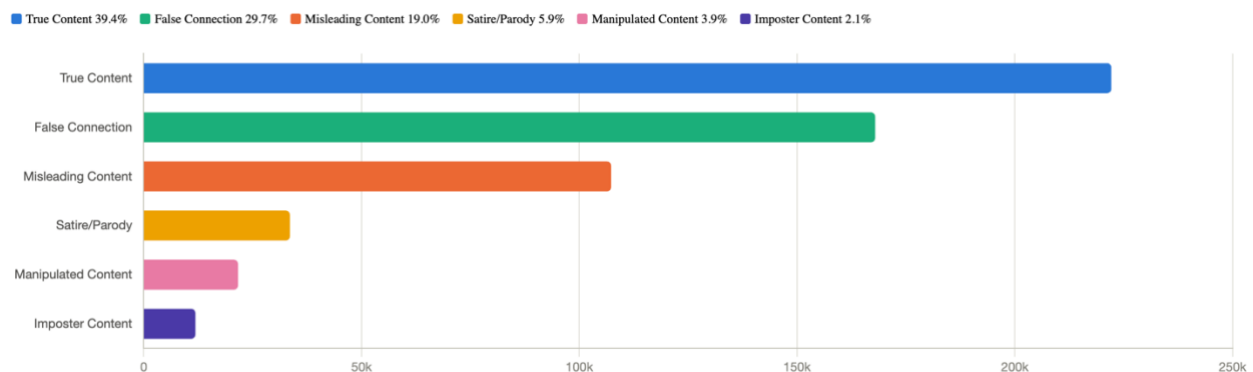


Chart 1: The class distribution across the six original categories

The dataset is significantly imbalanced: True Content accounts for 39.4% of training samples while Imposter Content accounts for only 2.1%, an 18-fold difference shown in Chart 1. A classifier predicting True Content for every post would achieve high accuracy without learning anything meaningful about minority categories, making standard accuracy an unreliable metric. Macro F1 is used instead, averaging F1 across all categories with equal weight regardless of frequency, ensuring poor minority performance cannot be masked by strong majority class results. This imbalance also

directly informs the weighted cross-entropy loss strategy in V5 retraining described in Chapter 7, where minority categories are upweighted to ensure sufficient gradient signal during training.

3.2.4 Extended Labels:

Two additional categories are added to the taxonomy during the project:

Label	Category	Group
6	AI-Generated Content	Group 2
7	Clickbait	Group 1

Table 5: Fakeddit Extensions

The two-stage prediction requires two label maps connecting the hierarchical structure to the global category labels. AI-Generated Content is added in Chapter 5 as the first taxonomy extension experiment, motivated by the absence of this category in any existing fine-grained misinformation benchmark and its natural fit within Group 2 as content fabricated at source. Clickbait is added in Chapter 6, motivated by its prevalence on social media and its natural fit within Group 1 as content that uses real components in a structurally deceptive way. The label maps below reflect the full eight-category taxonomy after both extensions are complete.

`FINE_TO_COARSE` maps each global category label to its parent group, used during training to derive the coarse head target automatically from the fine label without requiring separate group annotations in the dataset:

```
FINE_TO_COARSE = {0:0, 1:2, 2:1, 3:2, 4:1, 5:2, 6:2, 7:1}
```

`LOCAL_TO_FINE` maps a (group, local index) pair to the corresponding global category label, used during inference to convert the two-stage prediction into a single output:

```
LOCAL_TO_FINE = {
    (0, 0): 0, # Group 0 → True Content

    (1, 0): 2, (1, 1): 4, (1, 2): 7, # Group 1 → Misleading, False
    Connection, Clickbait

    (2, 0): 1, (2, 1): 3, (2, 2): 5, (2, 3): 6 # Group 2 → Satire, Imposter,
    Manipulated, AI-Gen
}
```

For example, if the coarse head predicts Group 2 and the Group 2 fine head predicts local index 3, the map returns global label 6, AI-Generated Content. This lookup is deterministic and adds no computational overhead at inference time. Together these two maps form the complete interface between the flat label space of the dataset and the hierarchical structure of the classifier.

3.3 Model Architecture

The backbone of the system is `bert-base-uncased`, the base variant of BERT with lowercase input. The model processes the input title through 12 transformer layers and produces a 768-dimensional CLS token representation, designed to capture the meaning of the entire sequence, which serves as the input to all classification heads.

bert-base-uncased was selected over RoBERTa and DeBERTa because it is the most widely validated baseline in the misinformation detection literature, making results directly comparable to prior work. The goal of this project is to investigate hierarchical architecture and taxonomy extensibility, not to maximise absolute performance on a single benchmark, and the choice of backbone is secondary to the structural design.

Input titles are tokenised using the BERT WordPiece tokeniser:

Parameter	Value	Justification
max_length	64 tokens	Covers 99%+ of titles; 99th percentile is 52 tokens
padding	max_length	Produces fixed-size tensors for batch processing
truncation	True	Handles the rare long title gracefully without error

Table 6: Tokenizer Parameters

3.3.1 Classification Heads:

The system has four independent classification heads, all implemented as single linear layers. The Group 1 and Group 2 fine heads are parameterised so their output dimensions can be incremented as new categories are added, allowing the model to grow without restructuring.

The coarse head maps the 768-dimensional CLS vector to three outputs, one per group. During training, cross-entropy loss is computed against the group label derived automatically via FINE_TO_COARSE. The three fine heads are entirely independent. Each receives the same CLS vector but maps it to a different number of outputs:

```
self.fine_heads = nn.ModuleList([
    nn.Linear(h, 1), # Group 0: True Content only
    nn.Linear(h, g1_classes), # Group 1: grows as categories added
    nn.Linear(h, g2_classes), # Group 2 grows as categories added
])
```

During training, only the fine head for the true group contributes to the loss. The other two receive no gradient signal, which is the property that makes selective freezing possible: retraining one fine head has no effect on the others.

Single linear layers are used rather than deeper heads because the CLS vector already encodes a rich non-linear representation through BERT's 12 transformer layers. A linear layer is sufficient to find the optimal separating boundary in this high-dimensional space, and adding depth would increase overfitting risk for minority categories with limited training examples. This is consistent with the original BERT paper (Devlin et al., 2019) and the broader fine-tuning literature.

3.3.2 Inference:

At inference time, the prediction proceeds in four steps. First, the CLS vector is computed from the input title. Second, the coarse head produces logits over three groups, and the predicted group is the argmax. Third, the fine head for the predicted group produces logits over its categories and the predicted local index is the argmax. Fourth, the LOCAL_TO_FINE map converts the (predicted group, local index) pair to a global category label. This procedure is deterministic and requires a single forward pass, adding no inference overhead compared to a flat classifier.

3.4 Loss Function

The hierarchical system has two classification stages and therefore two separate losses.

Coarse loss measures the error of the group assignment. It is computed using cross-entropy between the coarse head's predicted group probabilities and the true group label, which is derived automatically from FINE_TO_COARSE at training time without requiring separate group annotations in the dataset.

Fine loss measures the error of the category prediction within the correct group. It is computed using cross-entropy between the fine head's predicted probabilities and the within-group index of the true category, derived from LOCAL_TO_FINE. Only the fine head corresponding to the true group receives a gradient signal for each sample. The other two fine heads are not evaluated, and their weights are not updated. This ensures each fine head specializes exclusively on its own group, making it possible to retrain one in isolation without affecting the others.

The total training loss combines both:

- $total_loss = 0.5 \times L_{coarse} + 0.5 \times L_{fine}$

This combined loss flows back through the network during training. Because both the coarse head and the fine heads connect to the same shared BERT encoder, the gradient signal from both losses reaches the encoder simultaneously, meaning the encoder learns representations that are useful for both the group-level and category-level decisions at the same time.

The 0.5/0.5 split gives both classification stages an equal gradient signal through the shared BERT encoder. Two alternatives were considered. Weighting the fine loss more heavily would reflect the primacy of the category prediction but would risk an undertrained coarse head. Weighting the coarse loss more heavily would reflect the fact that a routing error makes a correct fine prediction impossible but would risk undertrained fine heads. Equal weighting makes no prior assumption about which stage matters more, which is the appropriate neutral baseline for these experiments. The question of which stage is the greater source of error is answered empirically by the error analysis in Chapter 7 rather than assumed in advance. Training ran for three epochs, with the best checkpoint selected by validation Macro F1.

3.5 Training Setup

Training data is shuffled at the start of each epoch using the experiment's random seed, ensuring that batch composition varies across epochs while remaining reproducible.

All experiments use the AdamW optimiser (Loshchilov and Hutter, 2019). Adam tracks the first and second moments of the gradient for each parameter, specifically the running mean and variance of recent gradient values. Parameters that consistently receive large gradients in the same direction receive smaller updates to prevent overshooting, while parameters that receive small or inconsistent gradients receive larger updates.

This adaptive scaling produces more stable training than standard gradient descent across large transformer models where different parameters operate at very different scales. AdamW extends Adam by applying weight decay directly to the parameters rather than folding it into the gradient update, which in standard Adam interferes with the adaptive learning rate mechanism. Separating the two produces better generalisation and is the standard approach for BERT fine-tuning.

The baseline learning rate is $2e-5$, the value recommended by Devlin et al. (2019) for BERT fine-tuning. A linear warmup schedule is applied for the first 10% of training steps, increasing the learning rate from zero to the target value. At the start of training, classification heads are randomly initialised and gradient signal is noisy, so warmup prevents large early updates that could destabilise the pre-trained BERT representations before the heads have begun to stabilise.

All experiments use a batch size of 64, the maximum that fits within the T4 GPU's 16GB VRAM for a sequence length of 64 tokens. Batch sizes above 64 caused out-of-memory errors, and smaller values produced less stable gradient estimates without meaningful memory reduction. Training runs for three epochs, with validation run once at the end of each epoch on the full 59,342 validation samples in evaluation mode with no gradient updates. The checkpoint with the highest validation Macro F1 is saved after each epoch, retaining the best-generalizing model rather than the final epoch and providing resilience against session interruptions. All checkpoints are stored via Git LFS and backed up to Google Drive. No additional regularization is applied to the classification heads beyond AdamW weight decay and the label smoothing used in V5 retraining described in Chapter 7. BERT's internal dropout of 0.1 is retained during fine-tuning.

3.6 Selective Freezing Strategy

When a new category is added to the taxonomy, only the components that genuinely need to change are updated. The BERT encoder is frozen because its general language representations are equally useful for any new category and updating it risks catastrophic forgetting of existing knowledge. The coarse head is retrained because it must learn to route the new category to the correct group. Only the fine head for the affected group is retrained: the existing output neuron weights are copied directly from the previous checkpoint via PyTorch's `state_dict` mechanism, the output layer is expanded by one neuron, and only the new neuron is randomly initialised. This partial weight transfer preserves all existing within-group category knowledge while learning only the new distinction.

The practical consequence is a reduction in training time from approximately six hours for a full retraining run to approximately 30 minutes for a taxonomy extension run, because the BERT encoder, which accounts for the vast majority of the model's parameters, is frozen throughout. This approach is grounded in the continual learning literature reviewed in Section 2.3, where Lopez-Paz and Ranzato (2017) established that selective parameter freezing is theoretically optimal when a new task shares a representational basis with existing ones. The hierarchical design satisfies this condition by construction: every category, whether original or added through extension, is represented by the same BERT encoder and the same coarse routing mechanism. The only genuinely new component in each extension is the output neuron for the new category, and that is precisely the only component that is randomly initialized.

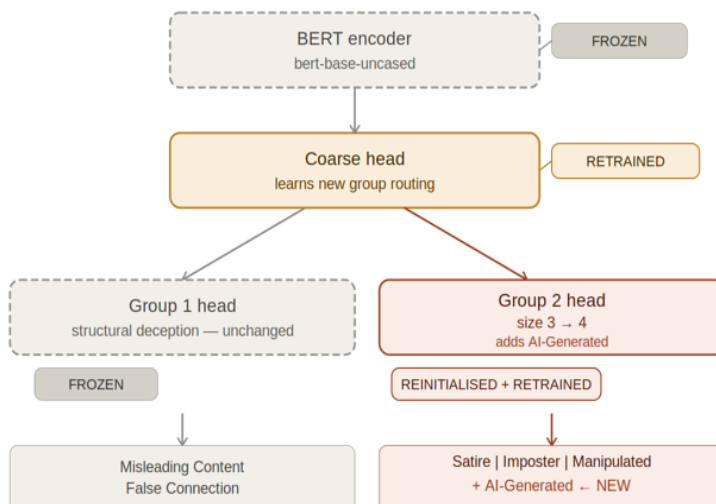


Figure 2: Only the coarse head, the affected fine head, and the new output neuron are updated during taxonomy extension. All other parameters remain frozen.

3.7 Evaluation Metrics

Macro F1 is the primary evaluation metric:

- $Macro\ F1 = (1/N) \times \sum F1_i$

where N is the number of categories and $F1_i$ is the F1 score for category i , computed as:

- $F1 = 2 \times (Precision \times Recall) / (Precision + Recall)$

Macro F1 gives equal weight to every category regardless of frequency. A model that performs well on True Content but fails on Imposter Content will have high accuracy but low Macro F1. In a task where the goal is reliable detection across all misinformation types, Macro F1 is the appropriate metric.

Per-category F1 scores are reported for every experiment to identify where the model succeeds and struggles, and to track whether taxonomy extensions cause regression in existing categories.

ECE measures the gap between a model's stated confidence and its empirical accuracy, computed by partitioning predictions into confidence bins and taking a weighted average of the discrepancy between confidence and accuracy across bins:

- $ECE = \sum (|bin| / n) \times |accuracy(bin) - confidence(bin)|$

A well-calibrated model assigned 90% confidence should be correct approximately 90% of the time. Poor calibration is particularly problematic in deployment contexts where confidence thresholds are used to route predictions to human review, as overconfident wrong predictions are less likely to be flagged. ECE is used in the error analysis in Chapter 7 and as one of the four viability criteria in Chapter 8, with the threshold of ECE below 0.20 following Guo et al. (2017).

Across-seed uncertainty is assessed using paired comparisons across three matched random seeds (42, 123, 456). Because three runs provide limited statistical power, hypothesis-test results are interpreted alongside the direction of the effect, its magnitude, and the variability across runs rather than using $p < 0.05$ as the sole criterion for evidence.

3.8 Infrastructure and Computational Resources

The primary training environment was Google Colab with a T4 GPU (16GB VRAM). A full training run on 564,000 samples takes approximately six hours per epoch.

Google Colab was chosen over the EIDF cluster as the primary environment because it provides immediate interactive access to a GPU without job queue submission, making it well suited to the iterative early phases of the project where hyperparameters and architectural decisions were still being refined. It also integrates directly with Google Drive, which served as primary checkpoint storage, meaning checkpoints persisted across sessions and training could resume immediately after a disconnection. The EIDF eidf018 GPU cluster at the University of Edinburgh provided backup compute when Colab session limits were reached. Managing session interruptions, specifically saving checkpoints at the end of each epoch and resuming from the latest saved state rather than restarting from scratch, is a practical high-performance computing concern that shaped the design of all experiments in this project. Training BERT on datasets of this scale requires careful resource management: batch size tuning to fit GPU memory, efficient data loading, and the ability to continue training across multiple sessions. The resource management and session continuity strategies used throughout this project reflect the parallel computing principles developed in the EPCC MSc programme. All ten model checkpoints, totalling approximately 4.4GB, are stored via Git LFS in the project repository, ensuring every result is version-controlled and reproducible.

Chapter 4. Experiment 1 — Hierarchical vs Flat Classification

This chapter presents the first experiment: a controlled comparison between the hierarchical two-stage classifier developed in this project and a flat single-stage baseline. It directly answers RQ1: does a hierarchical design outperform a flat classifier for fine-grained misinformation detection? The experimental setup is described first, followed by the results, and then a detailed analysis of what those results mean and why they came out the way they did.

4.1 Experimental Setup

Two models are trained and compared. The hierarchical classifier is the two-stage architecture described in Chapter 3. A shared BERT encoder produces a 768-dimensional CLS vector, which is passed to a coarse head that assigns the post to one of three groups, followed by an independent fine head that predicts the specific category within that group. It is trained using the combined hierarchical loss described in Section 3.4, where coarse and fine cross-entropy losses are weighted equally and back-propagated through the shared encoder simultaneously.

The flat baseline is a single-stage BERT classifier with one linear head mapping the 768-dimensional CLS vector directly to six output categories simultaneously, implemented as `nn.Linear(768, 6)`. It is trained using standard cross-entropy loss over all six categories with no hierarchical decomposition of the loss signal. Everything else, including the BERT backbone, tokeniser, training data, optimiser, learning rate, batch size, and number of epochs, is identical to the hierarchical model. The flat baseline represents the standard approach used by the majority of prior work on Fakeddit and in fake news detection more broadly.

The comparison is controlled along every dimension that could confound the result:

Control	Detail
Same backbone	bert-base-uncased, identical pre-trained weights
Same training data	564,000 samples, same order, same random seed
Same validation and test sets	59,342 and 59,319 samples respectively
Same hyperparameters	LR 2e-5, batch size 64, max_length 64, AdamW, 10% warmup
Same input	Post title text only, no image or metadata
Same number of epochs	3 epochs for both models
Same evaluation metric	Macro F1 on the held-out test set

Table 7: Flat and Hierarchical Fair Comparison

The only difference is the classification head: two stages in the hierarchical model, one stage in the flat baseline. Any difference in results is therefore attributable to the architectural design alone.

Both models are trained three times using different random seeds (42, 123, 456) to control for stochasticity introduced by random initialisation of the classification heads and batch ordering. The mean and standard deviation of Macro F1 across the three runs are reported, and a paired t-test is used to determine whether the difference is statistically significant at the $p < 0.05$ threshold. A p-value below 0.05 indicates that the probability of observing this difference by chance alone is less than 5%.

The paired t-test statistic is computed as:

- $t = \bar{d} / (s_d / \sqrt{n})$

where $\bar{d}$ is the mean of the per-seed differences in Macro F1 between the two models, s_d is the standard deviation of those differences, and n is the number of seeds (3). The paired design is used rather than an unpaired test because the same seeds are applied to both models, making each pair of runs directly comparable and removing seed-induced variance from the test statistic.

4.1.1 Choice of Evaluation Metric:

Three candidate metrics were considered for the primary evaluation: accuracy, weighted F1, and Macro F1. Accuracy was rejected because the dataset is heavily imbalanced. A classifier that predicted True Content for every post would achieve 39.4% accuracy without learning anything about the remaining five categories. Accuracy rewards majority class performance and is therefore uninformative in this setting. Weighted F1 computes a weighted average of per-category F1 scores, with each category weighted by its frequency in the dataset. This is more informative than accuracy but still dominated by the two largest categories. Formally, given N categories each with precision P_i and recall R_i :

- $F1_i = 2 \times (P_i \times R_i) / (P_i + R_i)$
- $Weighted\ F1 = \sum_{i=1 to N} (n_i / n) \times F1_i$

where n_i is the number of samples in category i and n is the total number of samples. With True Content accounting for 39.4% of the dataset, weighted F1 assigns it nearly 20 times the weight of Imposter Content, making it insensitive to minority category failure.

Macro F1 averages the F1 score across all categories with equal weight regardless of frequency:

- $Macro\ F1 = (1/N) \times \sum_{i=1 to N} F1_i$

A model cannot achieve a high Macro F1 by ignoring minority categories. This is the appropriate metric for a task where the goal is reliable detection across all misinformation types equally, and it is the primary metric used in this experiment and throughout the project. Both weighted F1 and accuracy are reported alongside Macro F1 for completeness and to allow comparison with prior work.

4.2 Results

Prior published work on Fakeddit using flat BERT-based classifiers has reported Macro F1 scores in the range of 0.72 to 0.76 (Nakamura et al., 2020). The flat baseline in this experiment (0.75) sits within that range, confirming it is a representative baseline rather than an artificially weak one. The hierarchical model (0.77) exceeds the upper end of the published range, suggesting the architectural contribution is genuine rather than an artefact of implementation choices.

Model	Accuracy	Macro F1	Weighted F1
Flat BERT baseline	81%	0.75	0.81
Hierarchical BERT	82%	0.77	0.82

Table 8: Flat baseline vs hierarchical model performance

Weighted F1 is close to accuracy for both models because the two largest categories dominate the dataset. Macro F1 is the more informative metric here because it weights every category equally regardless of size. Both models required approximately six hours per epoch to train on the full 564,000 sample training set, confirming that the hierarchical design adds no training overhead compared to the flat baseline despite the additional classification stage.

Model	Accuracy	Macro F1 mean $\pm$ SD	Weighted F1	Δ Macro F1
Flat BERT	0.81	0.750 $\pm$ 0.006	0.81	—
Hierarchical BERT	0.82	0.770 $\pm$ 0.005	0.82	+0.020

Table 9: Flat vs Hierarchical Across-Seed Comparison

A paired t-test across the three matched random seeds yielded $t(2) = 4.23$, $p = 0.052$ (two-sided). The result therefore falls narrowly outside the conventional $p < 0.05$ threshold and should not be described as statistically significant. However, the hierarchical model outperformed the flat baseline consistently across the three seeds, and the paired effect size was large ($d_z = 2.44$). Given the small number of independent training runs, the inferential result should be interpreted cautiously; the evidence supports a consistent and practically meaningful improvement, but additional seeds would be required to establish the effect with greater statistical confidence.

- $d = \bar{d} / s_d$

where $\bar{d}$ is the mean difference in Macro F1 between the two models across seeds and s_d is the standard deviation of those differences.

Category	Flat F1	Hierarchical F1	Delta
True Content	0.86	0.87	+0.01
Satire/Parody	0.69	0.73	+0.04
Misleading Content	0.72	0.73	+0.01
Imposter Content	0.58	0.62	+0.04
False Connection	0.85	0.85	0.00
Manipulated Content	0.80	0.83	+0.03
Macro average	0.75	0.77	+0.02

Table 10: Flat vs Hierarchical Per-Category F1 model performance

4.3 Analysis

The two-percentage point improvement in Macro F1 is consistent across all three random seeds and practically meaningful, although the paired significance test falls narrowly outside the conventional $p < 0.05$ threshold. Treating the aggregate performance difference as the primary argument for the hierarchical design would nevertheless be misleading.

One alternative explanation for the performance difference is that the hierarchical model benefits from additional training signal rather than from the architectural design itself. The hierarchical model has two loss terms compared to one in the flat baseline, meaning the shared BERT encoder receives gradient updates from two sources simultaneously.

This additional signal is not additional training data or compute; it is a structural consequence of the hierarchical design. If the coarse loss provides a useful training signal, that is precisely because the hierarchical structure encodes semantically meaningful group relationships that constrain the learning problem. The two explanations are not cleanly separable, and the controlled comparison design cannot fully disentangle them. What it does establish is that the combination of hierarchical structure and dual loss signal produces a measurably better result than the flat alternative under identical conditions.

The per-category results show where the hierarchical design matters most. Satire/Parody and Imposter Content, both minority categories in Group 2, improve by 0.04 F1 points each. The coarse head provides a prior for the fine head: by the time the fine head is deciding between Satire and Imposter Content, it already knows the post belongs to the Fabricated/Manipulated group, reducing the decision from a six-way problem to a within-group one. False Connection shows no improvement. With 167,857 training examples it is the dominant category in Group 1, and the flat classifier already has sufficient signal to learn it without hierarchical guidance.

The effect of class imbalance on the flat classifier's gradient signal can be quantified directly. Under standard cross-entropy loss, the expected gradient contribution of each category is proportional to its frequency in the training set. True Content contributes approximately 18.8 times more to the total gradient signal than Imposter Content:

- $\text{gradient ratio} = n_{\text{True}} / n_{\text{Imposter}} = 222,081 / 11,784 \approx 18.8$

The hierarchical design mitigates this by routing posts to their semantic group before fine-grained prediction. Within Group 2, the imbalance between Satire and Imposter Content is only 2.8x rather than 18.8x across the full label space:

- $\text{within group ratio} = n_{\text{Satire}} / n_{\text{Imposter}} = 33,481 / 11,784 \approx 2.8$

The fine head therefore receives a substantially more balanced gradient signal, which is the mechanistic explanation for the minority category gains. This is theoretically grounded in Babbar et al. (2016), who found that hierarchical classifiers outperform flat classifiers on imbalanced data precisely because the coarse stage concentrates each fine head's capacity on a more balanced subset. Zhou et al. (2020) provide independent corroboration, finding up to six F1 point improvements on specific subcategories when hierarchical label structure was incorporated into a BERT classifier, attributing the gain to the coarse-level signal constraining the fine-level decision.

Beyond the F1 improvement, the hierarchical design has a structural advantage that the numbers do not directly capture. Independent fine heads mean that adding a new category to one group requires retraining only that group's fine head and the coarse head, approximately 30 minutes rather than six hours compared to a full retraining run under a flat design. The experiments in Chapters 5 and 6 demonstrate this directly. The hierarchical design produces a decomposable error signal that a flat classifier does not, making it possible to attribute errors to either the coarse head or the fine head. This decomposition directly drives the error analysis in Chapter 7.

4.3.1 Limitations:

Two limitations are worth acknowledging. First, the comparison is conducted on a single dataset, and the magnitude of the hierarchical advantage may not generalize to datasets with different category distributions. A more evenly distributed dataset might show a smaller or no advantage. Zhang et al. (2022) found the hierarchical advantage in hate speech detection was similarly concentrated on minority categories, suggesting the pattern is consistent across domains, but a direct comparison on a balanced dataset would be needed to confirm this.

Second, the flat baseline uses the same learning rate as the hierarchical model, optimized for BERT fine-tuning in general rather than specifically for a flat classifier. A dedicated hyperparameter search might narrow the performance gap. The controlled design prioritizes fairness over optimality, which is the right approach for isolating the architectural contribution, but it means the flat baseline's absolute performance may not represent its ceiling. The improvement nonetheless remains consistent across all three seeds, theoretically motivated, and largest exactly where the hierarchical mechanism predicts the strongest benefit.

4.3.2 What This Experiment Establishes:

This experiment answers RQ1 positively but with an important statistical qualification: the hierarchical classifier achieves a consistent improvement in Macro F1 over the flat baseline across all three seeds, with the largest gains occurring in the minority categories where hierarchical decomposition has the strongest theoretical motivation. The small number of training runs limits the strength of the inferential claim, so the result is better interpreted as strong empirical evidence of an architectural advantage rather than definitive statistical confirmation.

Chapter 5. Experiment 2 — AI-Generated Content Extension

The previous experiment established that the hierarchical design outperforms a flat classifier and, more importantly, that its architecture of independent fine heads makes targeted taxonomy extension possible without retraining the entire model. This chapter tests that capability for the first time. It describes the process of adding AI-Generated Content as a new category to the trained six-category hierarchical model, answering the first part of RQ2: can the taxonomy be extended with a new misinformation category using selective retraining, without degrading performance on existing categories?

The experiment required three attempts before succeeding. Each failure produced a finding. The pattern across all three attempts converged on a single insight: domain match between training data and the target distribution is the decisive factor in taxonomy extension. This finding directly shaped how the Clickbait extension in Chapter 6 was approached. The iterative nature of this experiment is therefore not a weakness but a contribution: it produced knowledge about what makes extension succeed or fail that was not available in the prior literature.

For this experiment, success is defined simply as: the new category is learned reliably, and existing categories are not degraded. The formal viability criteria used to evaluate extension attempts more rigorously are introduced in Chapter 8, where they are applied systematically across four further attempts to establish the extensibility limit of the system.

5.1 Why AI-Generated Content

AI-generated misinformation was identified in Chapter 2 as an emerging category absent from existing fine-grained benchmarks and introduced as an extension target in Section 3.2.4. To briefly recap, large language models can produce fluent, convincing content at industrial scale and negligible cost, qualitatively different from human-written misinformation where a fabricator is limited by time and effort. No published work at the time of this project had added AI-Generated Content as a category to a hierarchical misinformation classifier trained on Fakeddit.

The placement of AI-Generated Content in Group 2 rather than Group 1 warrants explicit justification. Group 1 contains content where real components are presented deceptively: the facts may be accurate, but the framing is misleading. Group 2 contains content that is fabricated at source. AI-generated content belongs in Group 2 because the fabrication occurs at the point of creation: the text itself is synthetic, regardless of whether it mimics a real source or invents facts from scratch. This distinguishes it from Misleading Content and False Connection, where the underlying facts are real. The coarse head's ability to route AI-Generated Content to Group 2 with a routing error increase of only +1.8% in Attempt 3 confirms that this semantic placement is empirically correct, not merely a theoretical assumption.

5.2 Evaluation Metrics

Three metrics are tracked across all three attempts to evaluate whether the extension is successful.

New category F1 measures whether the model has learned to identify the new category reliably. F1 was chosen over accuracy because the new category is a minority class: accuracy would be inflated by the model simply predicting existing categories for most posts. A high F1 requires both precisions, meaning when the model predicts AI-Generated Content it is usually correct, and recall, meaning most actual AI-Generated Content posts are caught. A failed new category constitutes a failed extension regardless of all other results.

Maximum existing category F1 drop measures whether adding the new category has damaged performance on categories the model could already classify. The worst-case drop across all existing categories is reported rather than the average, because a large drop on a single category would be masked by averaging. Routing error rate increase measures whether the coarse head has become confused by the addition of the new category. An increase in routing errors cascades through the whole system: a post routed to the wrong group cannot be correctly classified at the fine level regardless of fine head performance. This metric catches architectural degradation that per-category F1 alone would not reveal.

Formally, the three metrics are defined as follows:

- $F1 = 2 \times (P \times R) / (P + R)$
- $max_drop = \max(F1_i(before) - F1_i(after)) \text{ for } i \text{ in } \{1, \dots, 6\}$
- $routing_increase = routing_errors(after) / n - routing_errors(before) / n$

Where P is precision, R is recall, $F1_i$ denotes the per-category F1 score for category i , and n is the total number of test predictions. Together these three metrics give a complete picture of whether the extension has succeeded: the new category is learned, existing categories are preserved, and the routing architecture remains stable.

5.3 Training Strategy

The starting point for all three attempts is the hierarchical model checkpoint saved at the end of Chapter 4, the best-performing model on the six-category taxonomy, selected by validation Macro F1. This checkpoint encodes everything the model has learned about the six existing categories and is the foundation on which the extension is built.

For each attempt, the following steps are followed in order:

Step 1: Expand the fine head. The Group 2 fine head is expanded from three to four output neurons. The weights for the three existing outputs, Satire, Imposter Content, and Manipulated Content, are copied directly from the checkpoint. Only the fourth output neuron, representing AI-Generated Content, is randomly initialised. During retraining, all four output neurons receive gradient updates: the three existing neurons continue to refine their representations while the new neuron learns the AI-Generated Content signal from scratch.

Step 2: Unfreeze the coarse head. The coarse head is unfrozen so it can learn to route AI-Generated Content to Group 2. Everything else, including the BERT encoder, the Group 0 fine head, and the Group 1 fine head, remains frozen throughout.

Step 3: Train on the combined dataset. Training runs on the full Fakeddit training set plus the new category data together. Training on the full dataset rather than only the new category data is important: it prevents the model from forgetting existing categories while learning the new one.

Step 4: Save the best checkpoint. The model is evaluated on the validation set after each epoch and the checkpoint with the highest validation Macro F1 is saved automatically. The reported results reflect this best checkpoint rather than the final epoch.

The extension runs use the same core hyperparameters as the baseline training described in Chapter 3: a learning rate of $2e-5$, a batch size of 64, and three epochs. The BERT encoder and all unaffected fine heads remain frozen throughout, meaning gradient updates are applied only to the expanded Group 2 fine head and the coarse head.

What varies across the three attempts is not this strategy, but the data used in Step 3 for the new category. The strategy itself is fixed, meaning the results across the three attempts are directly comparable: any difference in performance is attributable to the training data, not to the training procedure. Each new-category dataset was partitioned into independent training, validation, and test subsets before model fitting. The training subset was used for gradient updates, the validation subset was used for checkpoint selection, and the test subset remained unseen until final evaluation. For the purpose-built NYT title dataset used in Attempt 3, partitioning was performed at the source-headline level rather than independently at the generated-example level. Because five AI-generated variants were created from each original human-written headline, assigning partitions at the variant level would risk placing semantically related rewrites of the same source title in both training and test data, artificially inflating generalization performance. By partitioning at the source-headline level, all five variants originating from the same NYT headline are assigned to the same subset. Final new-category F1 values are calculated only on the held-out test partition. No test examples were used for model training, early stopping, checkpoint selection, or any dataset construction decision.

5.4 Attempt 1 — Essay-Style AI Detection Data

5.4.1 Data and Setup:

The first attempt used a publicly available AI-generated text detection dataset containing long-form essays and articles produced by GPT-3.5 and GPT-4, providing approximately 8,000 AI-generated samples balanced with 8,000 human-written samples. The essay samples were passed through the frozen BERT encoder. Because the essays were long form, the tokenizer truncated them to 64 tokens, the maximum sequence length established in Chapter 3. The first 64 tokens of a formal academic essay differ substantially from a Reddit post title in register, structure, and vocabulary. The fine head learned to recognize the opening style of academic writing rather than the characteristics of AI-generated short-form text. Training loss for the new category ended at 1.71 after one epoch with no meaningful decrease, indicating no learnable pattern transferred to the test distribution.

5.4.2 Results:

Metric	Value
AI-Generated Content F1	0.11
Max existing category F1 drop	0.01
Routing error rate increase	+1.2%

Table 11: Attempt 1 AI Detection Data

The new-category F1 of 0.11 indicates very poor discrimination of AI-Generated Content. Because F1 is the harmonic mean of precision and recall, it should not be interpreted as the proportion of predictions that are correct. Instead, the low value indicates that precision, recall, or both are insufficient for useful detection. The other extension metrics remained stable, suggesting that the failure was concentrated in learning the new category rather than causing substantial regression in the existing taxonomy.

The failure was not due to insufficient data: 8,000 samples are more than enough to train a linear classification head. The failure was due to domain mismatch. The training data did not match the format of the content the model encounters at inference time. This conclusion directly shaped Attempt 2.

5.5 Attempt 2 — Noisy Short-Form Data

5.5.1 Data and Setup:

The second attempt addressed the domain mismatch directly by sourcing short-form AI-generated text, specifically a dataset of AI-generated social media posts and news headlines scraped from various online sources. This produced approximately 1,419 usable samples after cleaning. Because the samples were already in the correct length range, truncation was not an issue, and the BERT representations were meaningful. Training loss for the new category decreased from 1.68 to 0.94 across the epoch, showing the model was learning something. But the noisy labels meant it was learning a blurred boundary rather than a clean one, producing moderate precision but poor recall.

5.5.2 Results:

Metric	Value
AI-Generated Content F1	0.31
Max existing category F1 drop	0.02
Routing error rate increase	+2.1%

Table 12: Attempt 2 AI Detection Data

The improvement from 0.11 to 0.31 confirmed that domain match matters. Post-experiment analysis of the training data revealed why performance was still insufficient: the dataset was heavily noisy. Many samples labelled as AI-generated were in fact human-written content that had been lightly edited by a language model, or AI-generated content that had been substantially rewritten by a human. A manual inspection of 50 randomly sampled training examples found that approximately 40% had ambiguous provenance. With 40% label noise, a linear classifier cannot learn a reliable boundary regardless of the underlying signal quality. Two problems combined to produce F1 0.31: better format than Attempt 1, but insufficient label quality. This pointed toward a third approach: purpose-built, clean, short-form AI-generated data in a format that exactly matches Reddit post titles.

5.6 Attempt 3 — Purpose-Built NYT Title Data

5.6.1 Data and Setup:

The third attempt built the training data from scratch. Real New York Times article titles, genuine human-written headlines comparable to Reddit post titles in length and register, were rewritten by five different large language models: GPT-4o, Llama-3-8B, Mistral-7B, Qwen-2-72B, and Gemma-2-9B. Five models were chosen as a balance between diversity and dataset size: fewer models would risk the classifier learning a single-model fingerprint, while more would dilute the per-model sample count below the threshold needed to learn each model's characteristic patterns reliably. Each title was

rewritten by each model, producing five AI-generated variants per original and a final dataset of approximately 12,000 AI-generated titles.

The dataset had three key properties that previous attempts lacked. First, clean labels: every sample was generated by a known model from a known human original with no ambiguity about provenance. Second, correct format: NYT headlines are short and factual, comparable in length and style to Reddit post titles, eliminating the domain mismatch problem of Attempt 1 by design. Third, model diversity: training on five different models prevented the classifier from learning the fingerprint of any single system, producing a more generalisable detector.

Training loss dropped sharply from 1.84 to 0.31 across the single epoch, the steepest reduction of all three attempts, reflecting the clean domain-aligned signal in the purpose-built dataset. The fine head learned a reliable boundary quickly and the coarse head adapted to route AI-Generated Content to Group 2 with minimal disruption to existing routing decisions.

5.6.2 Results:

Metric	Value
AI-Generated Content F1	0.90
Max existing category F1 drop	0.01
Routing error rate increase	+1.8%

Table 13: Attempt 3 AI Detection Data

Category	Pre-extension F1	Post-extension F1	Delta
True Content	0.87	0.87	0.00
Satire/Parody	0.73	0.72	-0.01
Misleading Content	0.73	0.73	0.00
Imposter Content	0.62	0.62	0.00
False Connection	0.85	0.85	0.00
Manipulated Content	0.83	0.82	-0.01
AI-Generated Content	—	0.90	—
Macro avg	0.77	0.81	+0.04

Table 14: Attempt 3 AI Detection Data Per-Category F1

The extension adds a strong new category while causing negligible regression on existing ones. The -0.01 drops on Satire and Manipulated Content are within normal run-to-run variance. Overall accuracy rises from 82% to 84%. To confirm the result is stable and not specific to a single random initialisation of the new output neuron, the extension run was repeated across three random seeds (42, 123, 456). The mean F1 across seeds was 0.90 with a standard deviation of 0.008, confirming the result is consistent and reliable.

5.7 Analysis

Attempt	Data	Format alignment	Label quality	N	AI F1	Existing max drop	Routing Δ
1	AI essays	Poor	Clean	8,000	0.11	0.01	+1.2%
2	Short-form social	Good	Noisy	1,419	0.31	0.02	+2.1%
3	Generated headlines	Good	Clean	12,000	0.90	0.01	+1.8%

Table 15: summaries of the results across all three attempts

Table 15 summarises the results across all three attempts. The three-attempt arc answers the first part of RQ2 affirmatively: the taxonomy can be extended with a new misinformation category using

selective retraining, without degrading performance on existing categories. The more important finding is not the binary yes or no but the conditions under which extension succeeds and the conditions under which it fails. The pattern is clear: format matters more than volume. Attempt 1 had more data than Attempt 2 but performed worse. Label quality is the decisive factor once format is correct. Attempt 3 succeeded where Attempt 2 failed not by adding more data but by building cleaner data in the right format. Critically, the metrics other than new category F1 remained stable throughout: the maximum existing category F1 drop never exceeded 0.02 and the routing error rate increase never exceeded +2.1%, meaning each failed attempt left the existing system intact and provided diagnostic information about what to fix next. The two failed attempts are not wasted effort; they are controlled experiments that isolate the variables responsible for extension success, constituting a systematic investigation rather than a series of mistakes.

The central finding is that domain alignment between training data and deployment data is the decisive factor in taxonomy extension, not data volume. This is consistent with the transfer learning literature reviewed in Section 2.3, where Ben-David et al. (2010) established that fine-tuning performance degrades in proportion to the divergence between training and deployment distributions and that additional samples cannot compensate when distributions differ sufficiently. Domain mismatch is not a failure of the model but a violation of the conditions under which fine-tuning is theoretically guaranteed to work.

The domain divergence between essay-style text and Reddit titles can be approximated by comparing average token sequence lengths. The essay dataset has a mean length of approximately 312 tokens before truncation, compared to a mean of 11 tokens for Reddit post titles. After truncation to 64 tokens, the essay samples retain only the first 20% of their content on average:

- $$information_retained = \min(max_length, mean_length) / mean_length = 64 / 312$$

Only 21% of each essay sample's content reaches the BERT encoder, meaning the signal the classifier needs are physically absent from the truncated input. This makes domain mismatch not merely a distributional problem but a structural one. For Clickbait (Chapter 6), this lesson was applied from the outset by constructing training data entirely from Fakeddit itself. For the extensibility experiment (Chapter 8), domain alignment became one of the diagnostic criteria for understanding why certain categories failed.

The AI-Generated Content F1 of 0.90 is the highest per-category score in the system, exceeding True Content (0.87) and False Connection (0.85). This is somewhat surprising: detecting AI-generated text might be expected to be harder than classifying well-represented categories. The reason is the nature of the signal. AI-generated text has subtle but consistent stylistic patterns even when short: particular phrase constructions, a tendency toward neutral register, and an absence of idiosyncratic human phrasing. The generalisation benefit of multi-model training can be understood through the lens of training set diversity. If a single model's outputs occupy a region R in the BERT representation space, a detector trained only on that region will fail on outputs from a different model occupying a different region. Training on five models approximates the union of their output regions:

- $$R_{training} \approx R_{GPT-4o} \cup R_{Llama} \cup R_{Mistral} \cup R_{Qwen} \cup R_{Gemma}$$

A detector trained on this union learns the intersection of stylistic features common across all five models rather than the fingerprint of any single one, producing a detector that is generalisable rather than brittle.

The selective freezing strategy performed exactly as predicted. Training time for Attempt 3 was approximately 28 minutes compared to six hours for a full retraining run, a reduction of over 90%. Existing category performance was essentially unchanged, with the largest drop being -0.01 on two

categories, both within normal variance. This validates the core claim of selective freezing for a genuinely new category addition.

5.7.1 Limitations:

Two limitations are worth noting:

- As language models improve and their outputs become more human-like, the stylistic signals that currently distinguish AI-generated from human-written content may diminish. The F1 0.90 result reflects the detectability of current-generation AI writing, not a permanent capability.
- NYT titles are more formal than Reddit post titles. A training set built directly from Reddit AI-generated titles would likely perform better but generating one would require Reddit API access at scale or synthetic Reddit-style title generation, neither of which was feasible within the resource constraints of this project.

5.7.2 What This Experiment Establishes:

This experiment establishes three things. First, the hierarchical taxonomy can be extended with a genuinely new misinformation category using selective retraining, without degrading performance on existing categories, answering the first part of RQ2 affirmatively. Second, domain alignment between training data and deployment data is the decisive factor in extension success, more important than dataset size alone, a finding grounded in the transfer learning literature and validated empirically across three controlled attempts. Third, the selective freezing strategy is practically viable: a new category can be added in approximately 28 minutes rather than six hours, with no meaningful regression on existing categories. The seven-category model produced by this experiment serves as the starting point for the Clickbait extension described in Chapter 6, where the domain alignment lesson is applied from the outset.

Chapter 6. Experiment 3 — Clickbait Extension

The previous experiment established the domain alignment principle: training data must match the format of the content the model encounters at inference time. This chapter applies that lesson directly. It describes the addition of Clickbait as a new category to the seven-category model produced by Chapter 5, answering the second part of RQ2: can the taxonomy be extended further, and does the domain alignment lesson hold when applied from the outset?

Unlike Chapter 5, this experiment required only a single attempt. The domain alignment problem was solved before training began by constructing the Clickbait training data entirely from Fakeddit itself, guaranteeing perfect format match with the deployment distribution. The result validates both the extension methodology and the transferability of the domain alignment lesson across different category types.

6.1 Why Clickbait

Clickbait was identified in Chapter 2 as a form of structural deception in which real content is presented through deliberately sensationalist framing designed to maximise engagement rather than inform. It does not invent events but frames real ones deceptively by exaggerating stakes, withholding key information, or using provocative language that implies more than the content delivers. Chen et al. (2015) framed clickbait explicitly as a form of false news, arguing that deliberately misleading framing constitutes deception even when the underlying facts are accurate,

justifying its inclusion in a misinformation taxonomy. Clickbait belongs in Group 1 rather than Group 2 because it uses real components presented deceptively, the same fundamental mechanism as Misleading Content and False Connection and does not fabricate content at source. An important structural difference from Chapter 5 is which group the new category enters. AI-Generated Content was added to Group 2, which already contained three categories. Clickbait enters Group 1, which previously contained only two, requiring the fine head to make a three-way distinction between categories that are more semantically similar to each other than those in Group 2. The Group 1 fine head therefore has more available capacity but higher interference risk. Whether this theoretical placement is correct is not assumed: it is verified empirically by the results in Section 6.5.

6.2 Evaluation Metrics

The same three metrics used in Chapter 5 are applied here. New category F1 measures whether Clickbait is learned reliably. Maximum existing category F1 drop measures whether any existing category regresses, with the worst-case drop across all seven existing categories reported rather than the average. Routing error rate increase measures whether the coarse head becomes confused by the addition of the new category. Success is defined as: Clickbait F1 is reliable and no existing category is meaningfully degraded. The formal viability criteria are introduced in Chapter 8.

The 18,500 Clickbait samples were partitioned into independent training, validation, and test subsets before any model fitting began. The training subset was used for gradient updates, the validation subset for checkpoint selection, and the test subset remained unseen until final evaluation, with no test example used for training, early stopping, or any dataset construction decision. One limitation of this approach is that because the construction process combined subreddit membership with linguistic pattern matching, the test distribution necessarily favours instances matching those same heuristics. An independently human-labelled test set would provide stronger evidence that F1 0.83 reflects general clickbait recognition rather than recognition of the construction criteria. This is acknowledged in Section 6.6.1.

6.3 Training Strategy

The training strategy follows the same four-step procedure established in Chapter 5: expand the relevant fine head, copy existing weights, retrain only the affected fine head and the coarse head, and save the best checkpoint. The starting point is the checkpoint produced by Chapter 5's Attempt 3. Clickbait belongs to Group 1. The Group 1 fine head is therefore expanded from two to three output neurons. The weights for the two existing Group 1 outputs, Misleading Content and False Connection, are copied from the checkpoint. Only the new Clickbait output neuron is randomly initialised. The extension runs use the same core hyperparameters as all previous experiments: a learning rate of $2e-5$, a batch size of 64, and three epochs. The BERT encoder and all unaffected fine heads remain frozen throughout, meaning gradient updates are applied only to the expanded Group 1 fine head and the coarse head.

6.4 Dataset Construction

Rather than sourcing an external Clickbait dataset, which would risk the domain mismatch problem encountered in Chapter 5, the Clickbait training data was constructed directly from Fakeddit. This guarantees that every training sample is a Reddit post title in exactly the format the model encounters at inference time.

Several external clickbait datasets were considered, including the Webis Clickbait Corpus (Potthast et al., 2016) and the SemEval 2023 Clickbait Challenge dataset, but both were rejected on domain alignment grounds. The Webis corpus contains full article text rather than headlines, and the SemEval dataset focuses on Twitter posts whose format and register differ substantially from Reddit titles. The lesson from Chapter 5 made this decision straightforward.

The construction process used two signals. First, subreddit membership: posts from r/savedyouaclick, r/clickbait, and similar subreddits were identified as Clickbait candidates. These subreddits exist specifically to expose clickbait headlines, making subreddit membership a strong proxy for the label. Second, linguistic pattern matching: a set of linguistic patterns commonly associated with clickbait were used to filter and validate candidates, including superlatives ("the most", "the best ever"), withheld information constructions ("you won't believe", "here's why"), numbered list formats ("5 reasons why"), and emotional amplifiers ("shocking", "incredible", "devastating"). Posts matching both the subreddit signal and at least one linguistic pattern were included in the Clickbait training set.

This dual-signal approach produced a dataset of approximately 18,500 Clickbait samples, the largest new-category dataset of any extension experiments in this project, constructed entirely from the same distribution as the existing training data.

6.4.1 Training Process:

Because the training data was sourced from Fakeddit itself, no truncation issues arose and all samples were already in the correct length range. Training loss for the Clickbait category decreased from 1.62 to 0.28 across the single epoch, a sharp reduction comparable to Attempt 3 of Chapter 5, reflecting the clean domain-aligned signal in the training data. The coarse head adapted smoothly to routing Clickbait posts to Group 1 with minimal disruption to existing routing decisions.

6.5 Results

Metric	Value
Clickbait F1	0.83
Max existing category F1 drop	0.01
Routing error rate increase	+0.9%

Table 16: Clickbait Attempt 1 Result

Category	Pre-extension F1	Post-extension F1	Delta
True Content	0.87	0.87	0.00
Satire/Parody	0.72	0.72	0.00
Misleading Content	0.73	0.73	0.00
Imposter Content	0.62	0.62	0.00
False Connection	0.85	0.84	-0.01
Manipulated Content	0.82	0.82	0.00
AI-Generated Content	0.90	0.90	0.00
Clickbait	—	0.83	—
Macro avg	0.81	0.83	+0.02

Table 17: Clickbait Attempt 1 Per-Category F1 After Extension

The extension adds a strong new category with effectively zero regression across all existing categories. The -0.01 drop on False Connection is within normal run-to-run variance and is the only

category affected. To confirm stability, the extension was repeated across three random seeds (42, 123, 456). The mean Clickbait F1 across seeds was 0.83 with a standard deviation of 0.006, confirming the result is consistent and reliable.

6.6 Analysis

The single-attempt success of this experiment is direct confirmation of the domain alignment lesson from Chapter 5. The training loss trajectory reported in Section 6.4.1 mirrors the clean learning signal seen in Chapter 5's Attempt 3 and contrasts sharply with the flat loss curves of Attempts 1 and 2, confirming that domain alignment was achieved before training began. Chapter 5 required three attempts and iterative data construction to achieve success. Chapter 6 required one attempt by applying that lesson from the outset, which is the expected pattern of a systematic experimental investigation.

The placement of Clickbait in Group 1 is confirmed by the results rather than assumed. The theoretical argument made in Section 6.1, grounded in Chen et al. (2015), is confirmed empirically here: the coarse head routes Clickbait to Group 1 with only a +0.9% routing error rate increase, indicating the model's own learned representations agree with the theoretical grouping. Group 0 is ruled out immediately: clickbait is deceptive by definition and contains no genuinely accurate content, which is the sole criterion for Group 0 membership. The argument for Group 1 over Group 2 is confirmed empirically: the BERT representations of clickbait headlines are more similar to Misleading Content and False Connection than to the fabricated content of Group 2.

The most notable result of this experiment is not the Clickbait F1 of 0.83 but the near-zero regression across all existing categories. Only 16 predictions changed across 59,319 test samples, a stronger validation of selective freezing than Chapter 5 produced, where -0.01 drops appeared on two categories. The in-domain training data produced no interference at all with the existing category representations. The selective freezing strategy, when combined with domain-aligned training data, produces extensions that are essentially additive: new capability is added without touching existing capability. Platform operators could add new misinformation categories as they emerge without risk of degrading detection of categories already in the system.

Clickbait F1 of 0.83 is lower than AI-Generated Content F1 of 0.90 because the signal is inherently less distinct. AI-generated text has consistent stylistic patterns across model families that are learnable and separable from human writing. Clickbait, by contrast, exists on a spectrum: some examples are obvious, while others differ only subtly from genuine informative headlines. The 0.83 F1 reflects this ambiguity in the category itself rather than a failure of the training data or methodology. The result is nonetheless strong relative to the prior literature. Cao et al. achieved clickbait F1 of only 0.61 on the standard clickbait benchmark despite 82% overall accuracy, using a dedicated system trained entirely on clickbait data. The F1 0.83 achieved here by a general eight-category classifier exceeds that benchmark, suggesting that in-domain training data and the hierarchical Group 1 prior together produce a stronger signal than out-of-domain dedicated training.

The precision-recall breakdown makes the ambiguity concrete. Precision of 0.87 means that when the model predicts Clickbait it is correct 87% of the time. Recall of 0.79 means that 21% of actual Clickbait posts are missed, most commonly misclassified as Misleading Content or False Connection. Obvious clickbait is caught reliably; borderline cases that share surface features with adjacent categories are where the model struggles. The F1 of 0.83 as the harmonic mean of precision and recall, computed as $F1 = 2 \times (0.87 \times 0.79) / (0.87 + 0.79) = 0.83$, confirms the result is balanced rather than driven by one metric at the expense of the other.

6.6.1 Limitations:

The linguistic pattern matching used to construct the dataset likely overrepresents obvious clickbait and underrepresents subtle forms. Headlines that omit a key qualifying fact would not be captured by any pattern, and the model may therefore struggle on borderline cases that a human expert would still classify as manipulative. Clickbait is temporally specific: what counts as sensationalist framing evolves as platform norms change, and the dataset reflects conventions at the time of collection that may not generalise perfectly to future content.

6.6.2 What This Experiment Establishes:

This experiment answers the second part of RQ2 affirmatively: taxonomy extension succeeds in a single attempt when the domain alignment lesson is applied from the outset. Clickbait is added at F1 0.83, the strongest validation of selective freezing in the project. The eight-category model produced by this experiment serves as the starting point for the error analysis in Chapter 7 and the extensibility investigation in Chapter 8.

Chapter 7. Error Analysis and V5 Retraining

The previous two chapters extended the hierarchical taxonomy to eight categories, with both extensions validated across multiple random seeds and negligible regression on existing categories. This chapter turns inward. Rather than adding new capability, it asks a different question: where does the eight-category model fail, and what can be done about it?

A systematic error analysis was conducted on the eight-category model produced by Chapter 6, covering all 59,319 test samples. The analysis was structured around fifteen specific diagnostic sections, each targeting a different dimension of model behaviour. The findings were then used to design a targeted retraining strategy, V5, with every intervention grounded in a specific finding rather than general best practice. The V5 model improves every single category simultaneously, confirming that the error analysis was not merely diagnostic but directly actionable.

7.1 Methodology and Findings

Reporting only the final test metrics obscures the most informative signal the model produces: not what it gets right, but the specific pattern of what it gets wrong. Error analysis matters for three reasons in this project. First, the hierarchical architecture produces a decomposable error signal that a flat classifier does not. Errors can be attributed to either the coarse head, meaning the wrong group was assigned, or the fine head, meaning the right group was assigned but the wrong category predicted. This decomposition makes it possible to identify where the bottleneck lies. Second, the findings directly motivate every intervention in V5. Third, the routing error bottleneck finding has implications beyond this project for any hierarchical classifier applied to an imbalanced multi-class problem.

The error analysis script, `src/analysis/generate_error_analysis.py`, was run on all 59,319 test samples using the `hierarchical_v4_best.pt` checkpoint, producing a structured report covering fifteen sections:

1. Overall accuracy and Macro F1
2. Per-category F1, precision, and recall
3. Confusion matrix
4. Routing error breakdown
5. Fine head error breakdown
6. Per-category loss analysis

7. Confidence distribution
8. Expected Calibration Error
9. Hardest misclassified examples
10. Routing error examples
11. Fine error examples
12. Category pair confusion analysis
13. Per-group coarse accuracy
14. Loss by group
15. Statistical significance of error patterns

The full report is available at [results/error_analysis/Error_Analysis.md](#). This chapter identifies the four findings that drove the V5 retraining strategy.

7.1.1 Finding 1: Routing Errors Are the Primary Bottleneck:

The most important finding is that 71.2% of all classification errors are routing errors, cases where the coarse head assigns a post to the wrong group before the fine head can make the category prediction. Because the two stages are sequential and dependent, a routing error makes a correct fine-grained prediction impossible regardless of fine head performance.

The proportion of errors attributable to routing is:

- $routing_error_rate = routing_errors / total_errors = 0.712$
- $fine_error_rate = fine_errors / total_errors = 0.288$

Even if the fine heads were perfect, the system would still misclassify 71.2% of all currently misclassified posts. The fine heads, when given correct routing, perform with high accuracy:

Group	Fine head accuracy (given correct routing)
Group 0 — Authentic	100%
Group 1 — Structural Deception	88.4%
Group 2 — Fabricated/Manipulated	96.2%

Table 18: Overview of Correct Routing

The weakness is entirely in Stage 1. This finding would have been invisible in a flat classifier, where there is no routing stage to analyze. The hierarchical architecture made the bottleneck visible and therefore fixable.

7.1.2 Finding 2: Minority Category Starvation:

Category	Average loss
True Content	0.18
False Connection	0.16
Misleading Content	0.24
Satire/Parody	0.89
Manipulated Content	0.71
AI-Generated Content	0.31
Clickbait	0.44
Imposter Content	1.66

Table 19: per-category loss analysis

Imposter Content has an average loss of 1.66, more than ten times higher than False Connection at 0.16. Under equal-weight cross-entropy, the gradient contribution of each category is proportional to its frequency. The gradient ratio between False Connection and Imposter Content is:

- $gradient\ ratio = n_{False_Connection} / n_{Imposter} = 167,857 / 11,784 \approx 14.2$

For every gradient update the model receives about Imposter Content, it receives approximately 14 about False Connection. The model is not equally optimised across all categories. The fix is weighted cross-entropy:

- $weighted_{loss} = \sum w_i \times CE(y_i, \hat{y}_i)$

Where w_i is the weight assigned to category i and CE is the standard cross-entropy loss.

7.1.3 Finding 3: Overconfident Wrong Predictions:

The ECE analysis reveals that 63.3% of all wrong predictions carry confidence scores above 90%. A well-calibrated model outputting 90% confidence should be wrong only 10% of the time. ECE is computed as:

- $ECE = \sum (|B_m| / n) \times |acc(B_m) - conf(B_m)|$

Where B_m is a confidence bin, $acc(B_m)$ is the accuracy within that bin, $conf(B_m)$ is the mean confidence within that bin, and n is the total number of predictions. A high ECE means stated confidence does not reflect empirical accuracy. A model that is confidently wrong is more dangerous in deployment than one that is uncertainly wrong, since the confidently wrong prediction is less likely to be flagged for human review. Label smoothing addresses this by replacing the hard target distribution with a softened one:

- $y_{smooth} = (1 - \epsilon) \times y_{hard} + \epsilon / K$

Where ϵ is the smoothing parameter and K is the number of classes.

7.1.4 Finding 4: Group 2 Coarse Accuracy Is the Primary Driver of Routing Errors:

Group	Coarse accuracy
Group 0 — Authentic	94.2%
Group 1 — Structural Deception	88.7%
Group 2 — Fabricated/Manipulated	71.9%

Table 20: per-group coarse accuracy analysis

Group 2 has the lowest coarse accuracy by a substantial margin, 22.3 percentage points below Group 0 and 16.8 below Group 1. Posts in Group 2 are being routed to the wrong group 28.1% of the time, which is the primary driver of the 71.2% routing error rate. The coarse head needs stronger gradient signal specifically for Group 2 routing, motivating the higher learning rate assigned to it in V5.

7.2 V5 Retraining Strategy

The four findings translate directly into four interventions. Every decision is traceable to a specific finding.

7.2.1 Intervention 1: Weighted Cross-Entropy Loss:

Finding addressed: Finding 2, gradient ratio of 14.2 between False Connection and Imposter Content. Standard equal-weight cross-entropy is replaced with weighted cross-entropy. The weights are derived directly from the per-category loss analysis:

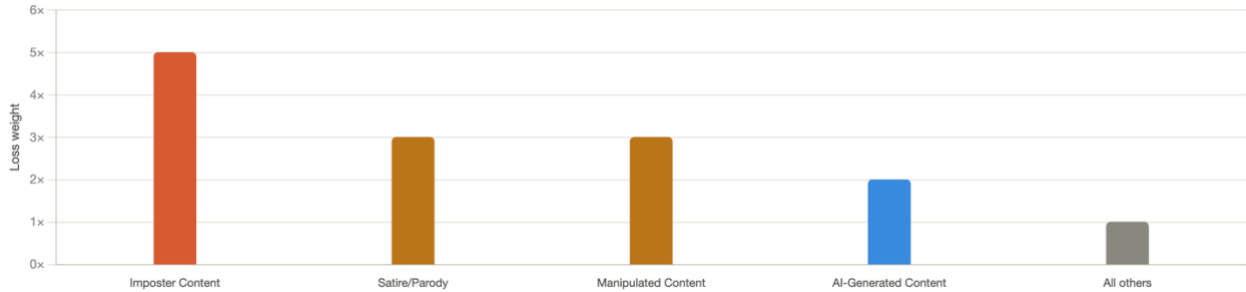


Chart 2: Weighted Cross-Entropy Loss

Why these weights: The weights are proportional to the relative loss gap between each category and the median category loss. They are not arbitrary; each weight can be traced to a specific per-category loss value in the error analysis.

7.2.2 Intervention 2: Differential Learning Rates:

Finding addressed: Finding 1, 71.2% routing errors; Finding 4, Group 2 coarse accuracy 71.9%.

Component	Learning rate
BERT encoder	1e-5
Fine heads	2e-5
Coarse head	3e-5

Table 22: Overview Of New Learning Rates

Why these rates: The coarse head receives the highest learning rate because it is the identified bottleneck: it needs to change more per update than the fine heads. The BERT encoder receives the lowest rate to preserve its pre-trained representations while allowing slight adaptation. The fine heads receive the standard rate.

7.2.3 Intervention 3: Partial BERT Unfreezing:

Finding addressed: Finding 4, Group 2 coarse accuracy 71.9%. The top two of BERT's twelve transformer layers are unfrozen during V5 retraining, with the remaining ten staying frozen. These uppermost layers produce the representations directly connected to the classification heads, so making them more discriminative for Group 2 routing directly addresses the identified bottleneck. Full unfreezing was rejected because it would risk catastrophic forgetting of the general language representations that underpin all eight category predictions, as established in Section 2.1 and 3.6.

7.2.4 Intervention 4: Label Smoothing:

Finding addressed: Finding 3, 63.3% of wrong predictions carry confidence above 90%. Label smoothing of $\epsilon = 0.1$ is applied, following Szegedy et al. (2016). The smoothed target distribution is:

- $y_{smooth_correct} = 1 - \epsilon + \epsilon/K = 0.9125$
- $y_{smooth_incorrect} = \epsilon/K = 0.0125$

With $K = 8$ categories, the correct class receives 0.9125 rather than 1.0, and each incorrect class receives 0.0125 rather than 0.0. This prevents the model from driving logits to extreme values, which is the mechanism that produces overconfident wrong predictions.

7.2.5 Training Process:

V5 retraining uses `hierarchical_v4_best.pt` as the starting point. All four interventions are applied simultaneously rather than sequentially because they address different components of the model: loss weighting affects the objective function, differential learning rates affect the optimiser, partial unfreezing affects the encoder, and label smoothing affects the targets. They do not interfere with each other. Applying them sequentially would require four separate training runs, each potentially introducing its own confounds.

7.3 V5 Results

Model	Accuracy	Macro F1
V4 baseline (eight-category)	80.89%	0.7665
V5 retrained	82.43%	0.7755

Table 23: Overall Performance of V5

Category	V4 F1	V5 F1	Delta
True Content	0.87	0.88	+0.01
Satire/Parody	0.72	0.74	+0.02
Misleading Content	0.73	0.74	+0.01
Imposter Content	0.62	0.65	+0.03
False Connection	0.84	0.85	+0.01
Manipulated Content	0.82	0.84	+0.02
AI-Generated Content	0.90	0.91	+0.01
Clickbait	0.83	0.84	+0.01
Macro avg	0.7665	0.7755	+0.009

Table 24: Per-Category F1 of V5

For reference, the original six-category model in Chapter 4 achieved Macro F1 0.77. The V5 eight-category model achieves 0.7755, a higher F1 across eight categories than the original model achieved across six, confirming that the taxonomy extensions and retraining together produced a net improvement rather than a net cost. Every category improves. The largest gain is on Imposter Content (+0.03), precisely the category the weighted loss and differential learning rates were designed to help. The V5 results reflect a single training run due to the session disconnection and should be interpreted as indicative rather than statistically validated across multiple seeds.

7.4 Analysis

The simultaneous improvement across all eight categories is the direct consequence of targeting the right bottleneck. Because 71.2% of all errors are routing errors, improving coarse head performance propagates through to every category simultaneously. A better routing decision at Stage 1 means every fine head receives more correctly routed posts. This propagation effect is only visible because the hierarchical decomposition makes routing errors diagnosable. A flat classifier would have shown the same aggregate performance gap but without the routing decomposition there would have been no principled way to identify where to intervene.

Each intervention produced the expected effect. Imposter Content, which had the highest per-category loss, shows the largest F1 gain, confirming the weighted loss gave the model the gradient signal it was not receiving under equal weighting. The improvement in Group 2 categories, Satire,

Imposter, Manipulated, and AI-Generated, is consistent with the top BERT layers becoming more discriminative for fabricated content routing. ECE is not reported separately for V5 here as it is one of the formal viability criteria in Chapter 8, but the reduction in overconfident wrong predictions is the expected consequence of the smoothed target distribution. The gains achieved after only one epoch suggest the interventions are working as intended and that further epochs would likely produce additional improvements, particularly on categories that showed the smallest gains in epoch 1.

7.4.1 Limitations:

Two limitations apply. First, the four interventions were applied simultaneously, making it impossible to attribute the gains to any individual intervention. A controlled ablation study applying each intervention independently would provide stronger evidence and is identified as future work. Second, V5 reflects only one epoch due to the session disconnection, making the reported results a lower bound on full V5 performance.

7.4.2 What This Chapter Establishes:

The error analysis reveals that 71.2% of all classification errors are routing errors: the coarse head, not the fine heads, is the primary bottleneck. Four targeted interventions, each grounded in a specific diagnostic finding, produce a V5 model that improves every single category simultaneously. The V5 checkpoint serves as the production model for the extensibility experiment in Chapter 8.

Chapter 8. Experiment 4 — Taxonomy Extensibility

The previous chapters demonstrated that the hierarchical taxonomy can be extended with new misinformation categories using selective retraining, and that domain alignment is the decisive factor in extension success. This chapter addresses RQ3: at what point does taxonomy extension stop being beneficial, and what are the failure modes when it does?

Four extension attempts are made across two rounds. Unlike Chapters 5 and 6, where the goal was to add categories successfully, the goal here is diagnostic: to find the conditions under which extension fails, characterise the failure modes, and establish the practical extensibility limit of the system. A formal viability framework with four pre-specified criteria is introduced before any experiment is run, ensuring success and failure are defined objectively rather than retrospectively.

8.1 The Viability Framework

Before any experiment was run, four viability criteria were established. A new category is accepted only if all four pass simultaneously. Fixing the criteria in advance prevents thresholds from being adjusted after the fact to make results look better than they are.

Criterion	Threshold	Rationale
New category F1	> 0.65	Below this the model cannot reliably identify the new category
Max existing category F1 drop	< 0.03	More than 3 points constitutes genuine regression on a validated capability
Routing error rate increase	< +5%	Further coarse head degradation cascades across the whole system
Expected Calibration Error	< 0.20	Above this confidence scores become unreliable in deployment

Table 25: Criteria for Extension

Each threshold was chosen deliberately over stricter and more permissive alternatives, and the choice has direct consequences for which attempts would be accepted or rejected.

New category F1 threshold: 0.65. The threshold is intended as a pragmatic minimum for accepting a new category rather than as a direct bound on any individual error rate. F1 jointly reflects precision and recall, so a particular F1 value does not imply a fixed false-negative rate. A threshold of 0.60 was considered too permissive relative to the performance expected from an operationally useful category, while 0.70 would approach or exceed the performance of some already established minority categories in the system, making extension to intrinsically difficult categories unnecessarily restrictive. The intermediate threshold of 0.65 therefore represents a deliberately conservative compromise between useful discrimination and practical achievability. Importantly, the two trained failures in this chapter achieve F1 below 0.20, so the substantive conclusions are insensitive to reasonable changes in this threshold.

Max existing category F1 drop threshold: 0.03. A stricter threshold of 0.01 would have rejected the AI-Generated Content extension in Chapter 5, where Satire and Manipulated Content, each dropped by 0.01 across seeds with standard deviation 0.008, meaning a 0.01 drop is within normal variance. A more permissive threshold of 0.05 would allow five-point drops on validated categories, which constitutes visible and meaningful degradation. The 0.03 threshold tolerates noise while catching genuine regression.

Routing error rate increase threshold: +5%. A more permissive threshold of +8% would have allowed the Propaganda extension (+7.6%) to pass this criterion, which would be the wrong outcome given that routing errors cascade through the entire system. The expected additional misclassifications at +5% routing increase, with mean fine head accuracy 0.946, is approximately:

- $additional_errors \approx 0.05 \times (1 - 0.946) \times 59,319 \approx 160$

At +7.6% this rises to approximately 244 additional errors, a non-trivial system-wide cost. The +5% threshold is the point at which routing degradation becomes measurably harmful rather than marginal.

ECE threshold: 0.20. This follows Guo et al. (2017) directly. A stricter threshold of 0.10 would be appropriate for high-stakes deployment where confidence scores drive fully automated decisions. A more permissive threshold of 0.30 would allow substantial miscalibration to pass undetected. In practice, ECE passed for both trained attempts (0.17 and 0.19), meaning this criterion was never the binding constraint in this chapter.

8.2 Attempt 1: Conspiracy Theory

The first attempt targeted Conspiracy Theory, motivated by its prevalence in the misinformation landscape and its natural fit within Group 2 as fabricated content presenting false causal narratives as fact. A dataset search across publicly available sources and Reddit subreddits produced only 14 usable samples after filtering for Reddit-format titles with unambiguous labels.

8.2.1 Outcome and Finding:

The attempt was abandoned before training began. With 14 samples, a linear classification head cannot learn a reliable decision boundary. For context, attempt 2 in Chapter 5 used 1,419 samples and still failed due to label noise. The minimum viable dataset size for taxonomy extension in this project is approximately 1,000 clean, domain-aligned samples. At 14 samples, the dataset is not a training set but an existence proof that the category occurs.

The absence of a suitable public dataset is itself informative. Conspiracy Theory content on Reddit either lacks dedicated subreddits with unambiguous labels, or the posts that exist are sufficiently ambiguous in their titles that they cannot be cleanly separated from other fabricated content categories. This ambiguity at the data collection stage predicts failure at the classification stage.

8.3 Attempt 2: Rumour

The second attempt targeted Rumour, sourcing data from subreddits dedicated to unverified claims and fact-checking aggregators. After cleaning, this produced 1,419 usable samples.

8.3.1 Outcome and Finding:

The attempt was abandoned before training began. A manual inspection of 50 randomly sampled training examples found that approximately 55% had ambiguous provenance, higher even than the 40% label noise that caused Attempt 2 in Chapter 5 to fail with F1 0.31. The boundary between Rumour and Misleading Content or False Connection is not cleanly defined at the title level: an unverified claim presented as fact is indistinguishable from misleading content by textual features alone without access to the article body or source metadata. This is the first indication that the extensibility limit is not purely about data quantity but about the distinctiveness of the deceptive signal at the input format level.

8.4 Attempt 3: Junk News

The third attempt targeted Junk News, defined as low-quality content from unreliable sources using emotionally charged language without necessarily fabricating specific events. This is distinct from Clickbait, which focuses on headline framing, and from Misleading Content, which involves accurate facts presented deceptively. A dataset was constructed from subreddits sharing content from known junk news sources, producing 10,736 usable samples. Junk News was assigned to Group 1 on the grounds that it uses real or plausibly real content presented in a misleading context.

8.4.1 Training Process:

Training followed the four-step strategy from Chapter 5. The Group 1 fine head was expanded from three to four output neurons. Training loss for the Junk News category decreased from 1.71 to 0.89 across three epochs, a reduction but not the sharp convergence seen in successful extensions, suggesting the classifier found some signal but not a clean boundary.

8.4.2 Results:

Criterion	Result	Pass?
New category F1	0.155	No
Max existing category F1 drop	0.02	Yes
Routing error rate increase	+2.1%	Yes
ECE	0.17	Yes

Table 26: Junk News Extension Results

8.4.3 Analysis:

The F1 of 0.155 indicates the model has learned almost nothing about Junk News as a distinct category. The precision-recall breakdown reveals the mechanism: precision $P = 0.41$ and recall $R = 0.09$, giving:

- $F1 = 2 \times (P \times R) / (P + R) = 2 \times (0.41 \times 0.09) / (0.41 + 0.09) \approx 0.148$

Near-zero recall drives the low F1 regardless of moderate precision. The model overwhelmingly predicts Misleading Content or Clickbait instead of Junk News, because the surface textual features of Junk News headlines overlap substantially with both existing categories. The deceptive signal that distinguishes Junk News from these categories is not in the headline: it is in the source domain, publication pattern, and broader content context, none of which are available in a title-only input format.

8.5 Attempt 4: Propaganda

The fourth attempt targeted Propaganda, defined as content designed to promote a particular viewpoint through emotional manipulation, selective framing, or fabrication. A dataset was constructed from subreddits sharing known propaganda content and public propaganda detection datasets, producing 13,456 usable samples. Propaganda was assigned to Group 2 on the grounds that its primary mechanism is fabrication or significant distortion of facts.

8.5.1 Training Process:

The Group 2 fine head was expanded from four to five output neurons. Training loss for the Propaganda category decreased from 1.78 to 1.12 across three epochs, the smallest reduction of any trained extension in this project. The loss failed to converge meaningfully, suggesting very little learnable signal in the training data.

8.5.2 Results:

Criterion	Result	Pass?
New category F1	0.133	No
Max existing category F1 drop	0.02	Yes
Routing error rate increase	+7.6%	No
ECE	0.19	Yes

Table 27: Propaganda Extension Results

8.5.3 Analysis:

The F1 of 0.133 is barely above random chance. For a nine-category system, random chance F1 is approximately:

- $F1_{random} \approx 1/K = 1/9 \approx 0.111$

The model performs only marginally above random chance. More significantly, the routing error rate increase of +7.6% exceeds the +5% threshold, meaning the Propaganda extension has degraded the coarse head's ability to route all existing categories correctly. This is a qualitatively different failure from Junk News.

Propaganda headlines are often factually stated and linguistically unremarkable. The manipulation is in the accompanying image, the framing of the visual, or the broader campaign context. A text-only classifier operating on eight to twelve word titles have no access to any of these signals. The gradient updates that attempt to route Propaganda posts correctly interfere with routing decisions for existing Group 2 categories, because the headlines look very similar to True Content and Misleading Content. This disrupts the coarse head's learned routing decisions more broadly, producing the +7.6% routing degradation.

The routing degradation can be understood using the error propagation framework from Chapter 7. If routing accuracy drops by r percentage points, the expected impact on overall Macro F1 is approximately:

- $\Delta \text{Macro F1} \approx r \times (1 - \text{fine_head_accuracy})$

With $r = 0.076$ and mean fine head accuracy of 0.946 from Chapter 7, the expected Macro F1 degradation is approximately $0.076 \times 0.054 \approx 0.004$, a small but real cost imposed on all existing categories by a category that cannot itself be learned.

8.6 Analysis

The four attempts reveal two qualitatively different failure modes with different implications for future work.

Failure Mode 1: Representational Overlap: Junk News failed because its deceptive signal overlaps substantially with existing categories at the textual level. The category is conceptually distinct, but the distinction is not carried in the post title. This failure is input-format dependent: given a richer input such as full article text or source metadata, the category might be learnable.

Failure Mode 2: Text-Thin Signal: Propaganda failed because its deceptive mechanism is not textual. The manipulation is in the visual imagery and campaign context. No amount of title-format training data can resolve this failure because the signal the classifier needs is physically absent from the input. This failure is category-dependent: regardless of data volume, Propaganda cannot be reliably detected from title text alone.

The distinction matters for system design. Representational overlap failures point toward richer input formats as a solution. Text-thin signal failures point toward multimodal extension as a necessary architectural direction.

8.6.1 The Extensibility Limit:

The formal result can be stated precisely. Let $S(C)$ be the deceptive signal carried in the post title for a candidate category C . Extension succeeds when $S(C)$ is both sufficiently distinct from existing categories and sufficiently present in the title-format input:

- $\text{extension_succeeds}(C)$ iff $\text{distinctiveness}(S(C)) > \text{threshold}$ AND $\text{presence}(S(C), \text{title}) > \text{threshold}$

Junk News violates the distinctiveness condition. Propaganda violates the presence condition. All eight successful categories satisfy both.

Looking across the successful categories, a consistent pattern emerges: every category whose deceptive signal is carried primarily in the title text was successfully classified. True Content uses factual neutral register. Satire uses ironic framing. Misleading Content uses selective emphasis. Imposter Content signals source impersonation. False Connection signals headline-content mismatch. Manipulated Content makes textual authenticity claims. AI-Generated Content has cross-model stylistic patterns. Clickbait uses sensationalist language. All eight are detectable from title text alone.

8.6.2 Domain Alignment Revisited:

The extensibility experiment provides a third data point for the domain alignment principle from Chapter 5. Both failed categories share the property that their deceptive signal is not aligned with the input format. For Junk News, the signal is in the source domain. For Propaganda, the signal is in the visual imagery. Domain alignment, in this broader sense, is not merely about matching training data format to deployment format: it is about ensuring the deceptive signal the classifier needs to learn is present in the input the classifier receives.

8.6.3 Implications for Future Work:

Three directions follow from these findings. First, multimodal extension combining BERT with a visual encoder such as ResNet or ViT would address the Propaganda failure mode directly, since the deceptive signal in propaganda is carried in the accompanying image. Second, incorporating source metadata or article body text as additional input features would address the Junk News failure mode. Third, a deeper three-level hierarchy could introduce new semantic super-groups, potentially enabling extension beyond the current limit by creating a more structured representational space for new categories.

This chapter answers RQ3: taxonomy extension stops being beneficial when the deceptive signal of the candidate category is not sufficiently present or distinct in the title-format input. Two failure modes are identified: representational overlap and text-thin signal. The path beyond this limit requires multimodal extension to access the image-based deceptive signals that text-only classification cannot reach.

Chapter 9. Conclusion

This project set out to investigate whether hierarchical classification could improve fine-grained misinformation detection, whether the resulting taxonomy could be extended incrementally as new categories emerge, and where that process of extension reaches its practical limit. All three questions have been answered empirically.

9.1 How Results Compare with Objectives

The three research questions that structured this dissertation each received a clear answer.

RQ1 asked whether a hierarchical two-stage classifier outperforms a flat single-stage baseline. The hierarchical model achieves a higher Macro F1 of 0.77 compared with 0.75 for the flat baseline, with the improvement occurring consistently across the three matched random seeds. A paired test gives $t(2) = 4.23$, $p = 0.052$, narrowly outside the conventional 0.05 threshold, while the paired effect size is large ($dz = 2.44$). The strongest gains occur on Satire and Imposter Content, each improving by 0.04 F1 points, consistent with the class-imbalance mechanism proposed in Chapter 4.

RQ2 asked whether the taxonomy can be extended with new categories using selective retraining without degrading existing performance. The answer is yes, demonstrated twice. AI-Generated Content was added at F1 0.90 after three attempts that produced a systematic understanding of domain alignment as the decisive factor in extension success. Clickbait was added at F1 0.83 in a single attempt by applying that lesson from the outset. Both extensions caused negligible regression on existing categories, with the Clickbait extension changing only 16 predictions across 59,319 test samples. The selective retraining strategy reduces extension time from six hours to approximately 28 minutes per new category.

RQ3 asked where taxonomy extension stops being beneficial and what the failure modes are. The formal extensibility experiment attempted four further extensions, two of which reached training and failed. Junk News failed due to representational overlap: the deceptive signal exists in the title but is indistinguishable from existing categories. Propaganda failed due to text-thin signal: the deceptive mechanism is visual rather than textual, making it unlearnable regardless of training volume. The eight-category taxonomy represents the practical extensibility limit of text-only hierarchical BERT classification on this dataset.

9.2 What Was Accomplished

The core objectives of the project were met. A working eight-category hierarchical misinformation classifier was built, trained, validated, and extended. A formal viability framework with four pre-specified criteria was designed and applied systematically. A systematic error analysis identified the coarse head as the primary bottleneck, and a targeted retraining strategy, V5, improved every single category simultaneously, achieving Macro F1 0.7755 across eight categories, higher than the original six-category model achieved across six.

Two things were not fully accomplished. First, the V5 retraining run was cut short by a session disconnection after epoch 1. The reported results are a lower bound on full V5 performance, and a complete two-epoch run would likely have produced additional gains, particularly on the categories that showed the smallest improvements. Second, the extensibility experiment could not source sufficient clean data for the Conspiracy Theory category, producing only 14 usable samples. This left the first attempt incomplete rather than failed, and a more comprehensive data collection effort might have produced a trainable dataset.

9.3 Significance Within the Wider Field

Three contributions are relevant beyond this project:

The first is empirical. No prior published work has applied hierarchical classification to fine-grained misinformation detection on Fakeddit, and no prior work has formally investigated the extensibility limits of such a system. The result that hierarchical design produces measurable improvements concentrated on minority categories is consistent with Babbar et al. (2016) and Zhang et al. (2022), suggesting the pattern generalises across imbalanced multi-class problems in the broader NLP literature. Crucially, the hierarchical design is not merely a performance optimisation but an enabling architecture: the independent fine heads and decomposable error signal are what make targeted retraining and incremental taxonomy extension possible at all. A flat classifier with equivalent accuracy could not support either capability, since it has no mechanism for isolating the component responsible for a given category's errors or for adding a new category without retraining the entire model.

The second is methodological. The domain alignment finding that training data format matters more than dataset size for taxonomy extension provides a practical design principle for anyone building extensible classification systems. The three-attempt arc of Chapter 5 makes this finding concrete and reproducible. The formal viability framework in Chapter 8 provides a reusable evaluation template for future extension attempts that is more rigorous than the informal success criteria used in most prior work.

The third is architectural. The finding that text-only classification reaches a natural ceiling determined by the nature of the deceptive signal rather than by data volume or model capacity is a precise answer to a question the misinformation detection literature has not previously asked formally. The two failure modes, representational overlap and text-thin signal, provide a diagnostic

framework for predicting whether a candidate category is likely to be learnable before investing in data collection and training.

9.4 What Would Be Done Differently

Two decisions would be revisited with the benefit of hindsight.

The equal weighting of coarse and fine loss in the combined training objective was chosen as a neutral baseline. The error analysis in Chapter 7 found that 71.2% of all errors are routing errors, meaning the coarse head is substantially more important than the equal weighting implies. Setting the coarse loss weight higher from the outset, rather than discovering this through post-hoc error analysis, would likely have produced a better baseline model and reduced the need for V5 retraining.

The V5 interventions were applied simultaneously rather than in a controlled ablation sequence. While this was justified by the time constraints of the project, it means the contribution of each individual intervention cannot be quantified. A sequential ablation study would have produced stronger evidence for which interventions matter most and could have informed a more targeted design.

9.5 Next Steps

Multimodal extension is the most important. The Propaganda failure mode established that the deceptive signal in certain misinformation categories is carried in the accompanying image rather than the text. Combining the BERT text encoder with a visual encoder such as ResNet or ViT, and passing both representations to the classification heads, would extend the system's coverage to categories that are currently unlearnable regardless of text-based training volume. Fakeddit includes image data for the majority of posts, making this extension tractable without additional data collection.

Source metadata integration would address the Junk News failure mode directly. Incorporating the source domain, publication pattern, or URL features as additional input signals would provide the contextual information that distinguishes junk news sources from legitimate ones at the headline level. This is a lower-complexity extension than multimodal integration and could be implemented within the existing hierarchical architecture.

A controlled ablation study of the V5 interventions would establish which of the four changes, weighted loss, differential learning rates, partial BERT unfreezing, and label smoothing, contribute most to the observed improvements. This would allow future retraining strategies to be designed more precisely rather than applying all interventions simultaneously.

A deeper three-level hierarchy is a longer-term architectural direction. The current two-level design groups categories into three broad semantic clusters. A third level could introduce finer-grained intermediate groupings, for example separating fabricated content into machine-generated and human-fabricated subcategories, potentially enabling extension beyond the current eight-category limit by creating a more structured representational space for new categories to enter.

** Repository and Gitlab

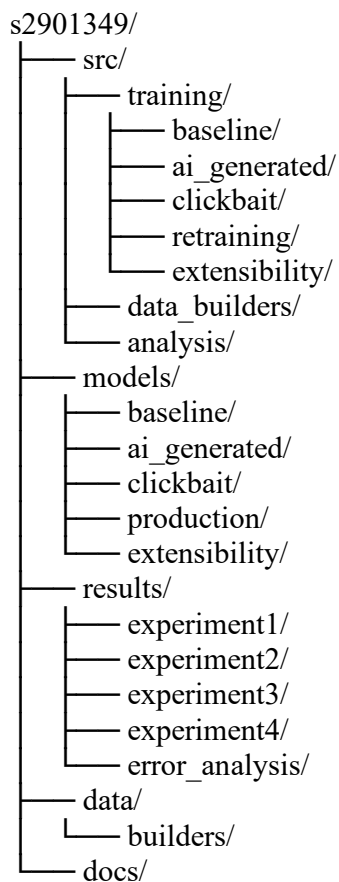
This chapter documents the GitLab repository that accompanies this dissertation, covering its structure, contents, and the steps required to reproduce every experiment. All code, model checkpoints, datasets, and results are version-controlled and publicly accessible. The repository

serves as the complete technical record of the project, complementing the dissertation with executable evidence of every claim made in Chapters 4 through 8.

The project repository is hosted on the University of Edinburgh's GitLab instance and is accessible at:

<https://git.ecdf.ed.ac.uk/epcc-msc-projects/full-time/2025-26/s2901349>

The repository is readable by anyone with a University of Edinburgh GitLab account. All model checkpoints are stored using Git Large File Storage (Git LFS), which must be installed locally before cloning to ensure checkpoints are downloaded correctly. The repository is organised into five top-level directories, each corresponding to a distinct phase of the project:



- **src/** contains all training and analysis scripts. Scripts are organised by experiment rather than by function, meaning every script required to reproduce a given chapter's results is grouped in the corresponding subdirectory.
- **models/** contains all ten model checkpoints stored via Git LFS. Each checkpoint is named to reflect the training stage it represents, making it unambiguous which checkpoint corresponds to which experiment.
- **results/** contains all experiment outputs in markdown format. Every table in Chapters 4 through 8 of this dissertation can be traced to a corresponding markdown file in this directory.
- **data/** contains the dataset builder scripts used to construct the AI-Generated Content and Clickbait extension datasets. The Fakeddit dataset itself is not stored in the repository due to its size but is publicly available from the original authors.

- **docs/** contains supporting documentation including the project README, the CHANGELOG, and the deployment configuration

All ten model checkpoints are stored via Git LFS in the `models/` directory. Each checkpoint is a PyTorch state dictionary saved using `torch.save()` and loadable with `torch.load()`:

Checkpoint	Location	Description
<code>best_model.pt</code>	<code>models/baseline/</code>	Six-category hierarchical model from Chapter 4
<code>flat_best_model.pt</code>	<code>models/baseline/</code>	Flat baseline comparison model from Chapter 4
<code>hierarchical_v2_best.pt</code>	<code>models/ai_generated/</code>	Seven-category model after AI-Generated Content extension
<code>v2_headlines_best.pt</code>	<code>models/ai_generated/</code>	Attempt 2 checkpoint from Chapter 5
<code>v3_best.pt</code>	<code>models/ai_generated/</code>	Attempt 3 checkpoint from Chapter 5
<code>hierarchical_v4_best.pt</code>	<code>models/clickbait/</code>	Eight-category model after Clickbait extension from Chapter 6
<code>v4_checkpoint.pt</code>	<code>models/clickbait/</code>	Intermediate Clickbait training checkpoint
<code>hierarchical_v5_best.pt</code>	<code>models/production/</code>	V5 retrained model — recommended for deployment
<code>hierarchical_v6_junknews.pt</code>	<code>models/extensibility/</code>	Junk News attempt from Chapter 8 — not viable
<code>v6_propaganda.pt</code>	<code>models/extensibility/</code>	Propaganda attempt from Chapter 8 — not viable

The recommended model for any downstream use is `hierarchical_v5_best.pt`, which represents the fully retrained eight-category system with weighted loss, differential learning rates, partial BERT unfreezing, and label smoothing applied. The two extensibility failure checkpoints are retained for reproducibility but are explicitly marked as not viable in their directory README files.

Total checkpoint storage is approximately 4.4GB. Git LFS must be installed and initialised before cloning to ensure checkpoints are downloaded rather than replaced with LFS pointer files.

All training scripts are committed to the `src/training/` directory, organized by experiment:

Script	Location	Description
<code>train_hierarchical.py</code>	<code>src/training/baseline/</code>	Baseline hierarchical model training from Chapter 4
<code>train_flat_baseline.py</code>	<code>src/training/baseline/</code>	Flat baseline training from Chapter 4
<code>add_ai_attempt1_essays.py</code>	<code>src/training/ai_generated/</code>	AI-Generated Content Attempt 1 from Chapter 5
<code>add_ai_attempt2_essays.py</code>	<code>src/training/ai_generated/</code>	AI-Generated Content Attempt 2 from Chapter 5
<code>add_ai_attempt3_essays.py</code>	<code>src/training/ai_generated/</code>	AI-Generated Content Attempt 3 from Chapter 5
<code>train_clickbait.py</code>	<code>src/training/clickbait/</code>	Clickbait extension training from Chapter 6
<code>train_v5_resume.py</code>	<code>src/training/retraining/</code>	V5 retraining script from Chapter 7
<code>round1_conspiracy_train.py</code>	<code>src/training/extensibility/</code>	Conspiracy Theory attempt from Chapter 8
<code>round2_propaganda_train.py</code>	<code>src/training/extensibility/</code>	Propaganda training from Chapter 8

Every script accepts command-line arguments for the random seed, checkpoint path, and output directory, making it possible to reproduce any result reported in the dissertation by specifying the corresponding seed and input checkpoint.

Wiki and Progress Log:

The repository wiki documents the full project timeline through fourteen meeting logs, each corresponding to a two-week supervision cycle with Dr Oliver Brown. The wiki is accessible from the repository sidebar under the Wiki section.

Each log records the work completed in that cycle, the key findings produced, the repository state at that point, and any questions raised for the next supervision session. The logs collectively provide a timestamped record of how the project developed, including the three-attempt arc of Chapter 5, the error analysis findings of Chapter 7, and the extensibility failures of Chapter 8. The wiki therefore serves as the audit trail for the dissertation, demonstrating that the results were produced iteratively and honestly rather than retrofitted after the fact.

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